RTA-SUM Partial Networking Coordinator (PNC)

Design Document

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1 Introduction and Goals

This model design document describes the system design of the model of the RTA-SUM module Partial Network Coordinator. The Standard Utility Module, Partial Network Controller (SUM_PNC), is a software component located in the Application Layer of the standardized architecture of AUTOSAR.

1.1 Introduction

1.1.1 System Purpose

SUM_PNC manages all of the BSW services related to Partial Networks (PNs) for the ECU that is a member of any PN. SUM_PNC shall be implemented in all ECUs that are members of any PN.
SUM_PNC is responsible for the following functions:

1. For ECUs that are Activating ECUs, SUM_PNC contains the PN Activation Criteria and Deactivation Criteria per the "Global B PNC Activation Specification". This document defines the Global B Partial Networks, roles of the ECUs within the Partial Networks, and activation/deactivation criteria for each Partial Network. Refer (*);
2. SUM_PNC is the only SW-C that interfaces with the BSW services to activate and deactivate Partial Networks;
3. For all ECUs that are Activated ECUs, SUM_PNC interfaces with the BSW services to track the activation status (active or inactive) of the PN;
4. For all ECUs that are members of any PN, SUM_PNC provides a port that is used to indicate communication capability from SUM_PNC for each SW-C;
5. Collection of all input signals and input integrity signals (e.g. Rx_MonitoredSignalAvailability from SUM_SSM) that are relevant to Activation/Deactivation Criteria for the PN in which the ECU participates. (Integrity signals apply to both the hardware and communication based signals.);
6. Collection of all conditional logic related to activation/deactivation criteria for the PN in which the ECU activates/deactivates;
7. Execute a sequence to fulfil the activation of a PN request;
8. Execute a sequence to fulfil the deactivation of a PN request;
9. Identify PN mapping to communication channel(s);

(*): The ECU's within the Partial networks are assigned with role of Activator ECU's and Activated ECU's. There will be set of conditions based on which the activators will set activation criteria/deactivation criteria.
Activation Criteria: a set of conditions that result in the activation of a PN. These condition are typically a combination of local inputs and / or serial data signals.
Deactivation Criteria: a set of conditions that result in the deactivation of a PN.

1.1.2 Document Purpose

This document introduces the key ideas of the architecture of Standard utility module: Partial Network Coordinator (SUM_PNC). Its intended audience comprises System Architects, Product Architects, Developers and Testers who need a rough understanding on the concepts of the product.

The document describes the product and its functionality on the component level, the interfaces and the interactions of the partial network coordinator with the rest of the standard utility modules, software components and BSW modules.

The document does not explain the components design in detail.

1.2 Goals of the Architecture

1.2.1 Goals

The main goals of the architecture are:

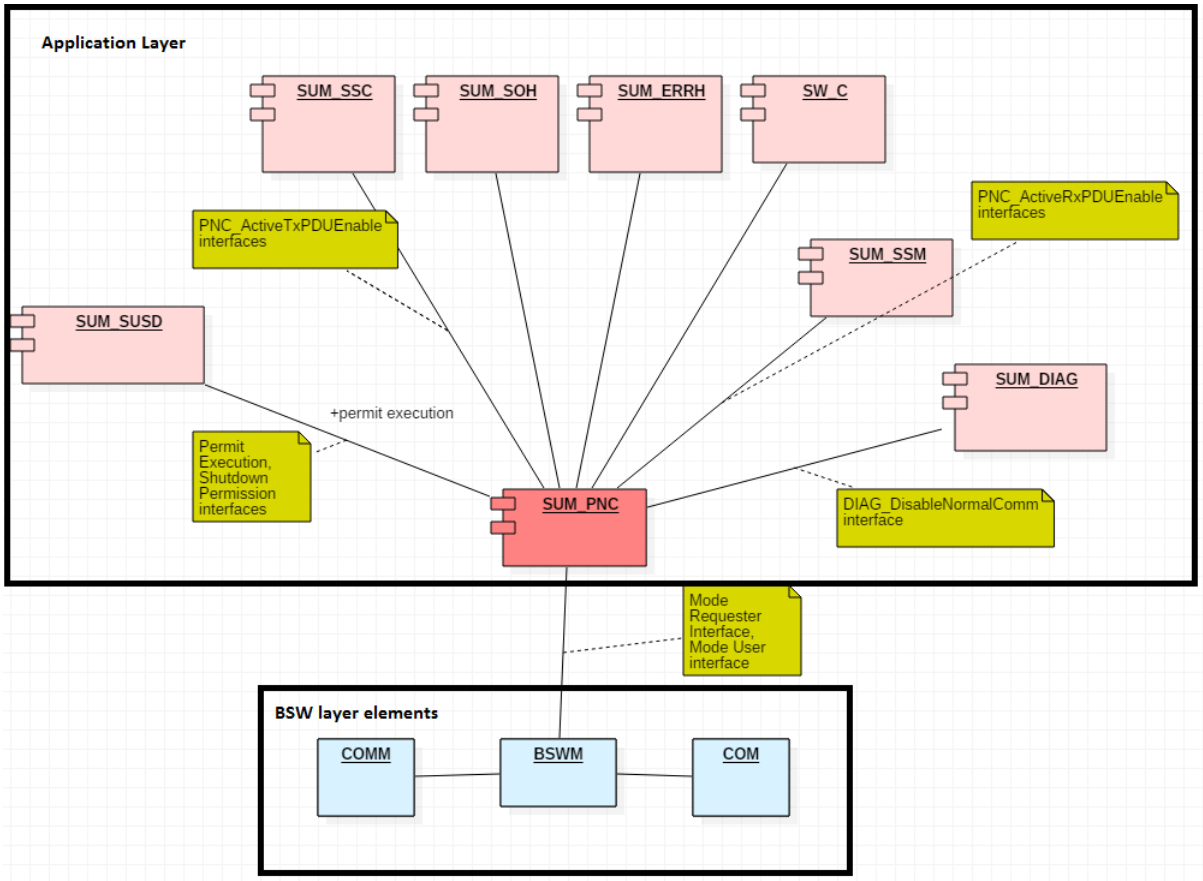
1. **Configurability**: The module has to be designed with the highest possible level of configurability, in order to be easy to adapt it in the future to changes on the standard or even in some project requirements;
2. **Maintainability**: The architecture of the component shall be clearly defined and easy to understand, in order to guarantee the maximum possible level of maintainability;
3. **Performance**: The component shall be designed in order to maximize run-time execution speed as well as memory consumption;
4. **Correctness of Implementation**: The Detailed Design must to be aligned to the Architecture document;

Maintainability

1. SUM_PNC shall implement a Mode User port interface to BswM to obtain notification of Partial Network status of all PNs which the ECU participates in.
2. SUM_PNC shall implement <n> port(s), PNC_ActiveTxPDUEnable<n> to publish notification of transmit ready capability for a specific Partial Network to SUM_SUSD and SUM_SSC.
3. SUM_ERRH, SUM_SOH, and SUM_SSC all require the PNC_ActiveTxPDUEnable interface.
4. All ECUs that are designated to process Partial Network activation and/or deactivation criteria shall implement a Mode Request port interface from SUM_PNC to BswM for the purpose of activating and/or deactivating Partial Networks relevant to the ECU.
5. All ECUs that are members of a Partial Network shall implement a Mode Request port interface from SUM_PNC to BswM for the purpose of activation and deactivating PDU Rx groups relevant to these Partial Networks.
6. All ECUs that are members of a Partial Network shall implement a Mode Request port interface from SUM_PNC to BswM for the purpose of activation and deactivating PDU Tx groups relevant to these Partial Networks.
7. SUM_PNC shall implement a port, Shutdown Permission, which can publish notification to SUM_SUSD to control the disable or enable shutting down the ECU.

1.3 Stakeholders

Below diagram illustrates SUM-PNC dependency with other SUMs and AUTOSAR System Services



Important stakeholders for this architecture are:

- SUM Modules: SUSD, SSC, SOH, ERRH, SSM, DIAG;
- BSW Modules: BswM, Com, ComM;

Following sections will help in understanding each SUM module and BSW module interfacing dependency and functionalities in detail.

1.4 Definitions, Acronyms and Abbreviations

Definition	Description
Activating ECU	ECU assigned to monitor and process PN Activation Criteria. At least one ECU per PN is an Activating ECU, but there may be others (depends on the architecture deployment)
Activated ECU	ECU is a member of an activated PN. All members of networks supporting PNs are capable of being Activated ECUs

Definition	Description
Activation Criteria	A set of conditions that result in the activation of a PN. These conditions are typically a combination of local inputs and / or serial data signals
API	Application Program Interface
AMM	Application Mode Manager
BSW	Basic Software
BswM	Basic Software Mode Manager
CAN	Controller Area Network
Com	Communication Module
Dcm	Diagnostic Communication Module
Deactivation Criteria	A set of conditions that result in the deactivation of a PN
DTC	Diagnostic Trouble Code
DIAG	Diagnostic Application Services
DFD	Data Flow Diagram
ECU	Electronic Control Unit
Ecuc	ECU Configuration File
ERRH	Error Handling Module
FSM	Finite State Machine
GM	General Motors
Locally Active PN	PN that is activated from within the ECU because its PN Activation Criteria was met
PNC	Partial Network Coordinator
Remotely Active PN	PN that is activated from outside the ECU
SOH	State Of Health Manager
SSM	Signal Status Monitor
SSC	Security Service Coordinator
SUM	Standard Utility Module
SUSD	Startup Shut down Coordinator
SW-C	Software Component
SW or S/W	Software
VMM	Vehicle Mode Manager
WRT/wrt	With respect to

1.5 Referenced Documents

Document Name	Description	Version	Release Date
GB8002 AUTOSAR Standard Utility Modules Specification	PNC module specification	24.24.158	01/08/2021
GB5280 Global B Partial Network Deployment Specification	Partial Network deployment specification	MY24.161R1	28/07/2021
GB6001 Diagnostic Infrastructure Specification	PNC DIDs specification	24.24.178.1p0	10/09/2024

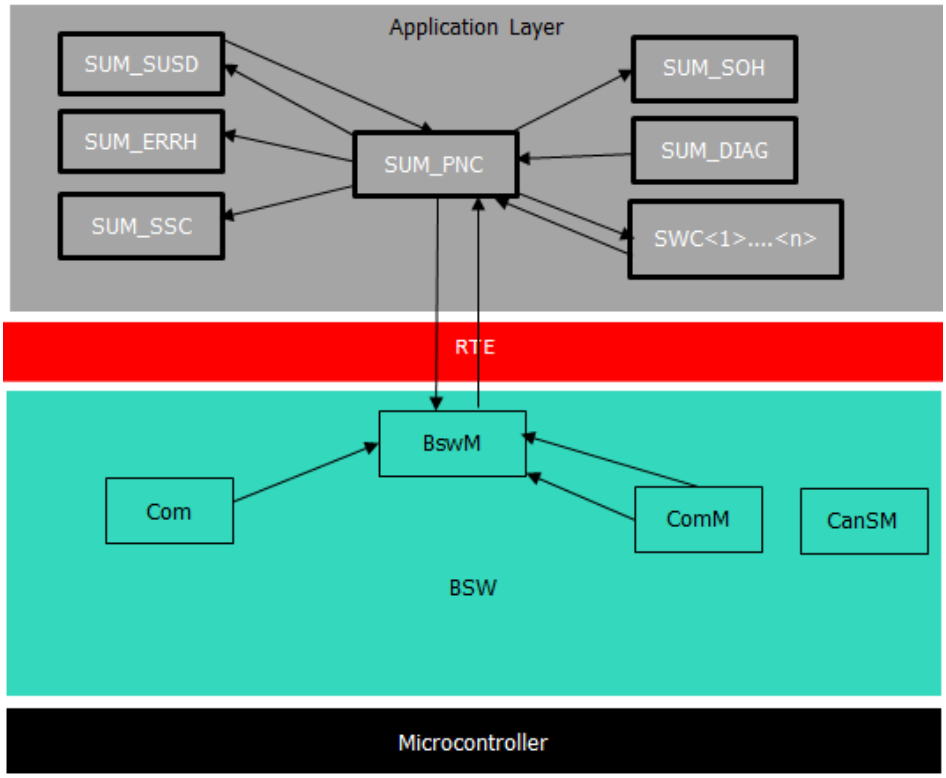
2 Architecture Constraints

2.1 Environment

Tool Name	Version No.	Description
RTA-CAR	12.3.3pr8	AUTOSAR authoring tool required for RTE,BSW and PNC configuration
RTA-RTE	12.4.0	RTE generator used for testing
CANTATA	9.5	Used for unit testing
PRQA Framework	2.4.0	Used for static analysis

2.2 Layers Definition

The Standard Utility Module, partial network Coordinator (SUM_PNC), is a software composition located in the Application Layer of the standardized architecture of AUTOSAR.



For detailed information on it refer to Autosar.org website. Follows a short description of layers definition.

2.2.1 Application Layer

Autosar Application layer consists of application software components, SUMs that are interdependent on each other for functional needs. PNC interacts with other SUM modules in the application layer, the importance of interaction with those SUMs are explained briefly below:

SUSD

SUSD provides signal to enable execution of PNC. Based on PN activity, PNC notifies this SUM for Shutdown permission.

SOH

Interface to publish PN<n> failed attempt to start.

SSC, ERRH, SOH

Interfaces to publish Tx status notification (enabled/disabled).

DIAG

DIAG Interface to PNC Provides notification to SUM_PNC the status of diagnostics enabling or disabling of normal communications for each channel.

SWC<1>...<n>

PNC interfaces to other SWCs to interface to Activation Criteria Signals:

- <ActivationSensor_n> - signals that are read by a Sensor SW-C that is local to the ECU
- <ActivationSignal_n> - signals that are received via serial data communications from a remote ECU. (These signals must be received via a different PN that the one which needs activation)

2.2.2 RTE Layer

Specifies the system function bus, main communication model etc.

2.2.3 BSW Layer

Provides BswM services via mode requester and mode user interfaces also services of communication stack are required by PNC.

2.3 Generic Component Interface

Below table summarizes the interfaces associated with PNC and dependent/required modules.

[$SDD_PNC 001]

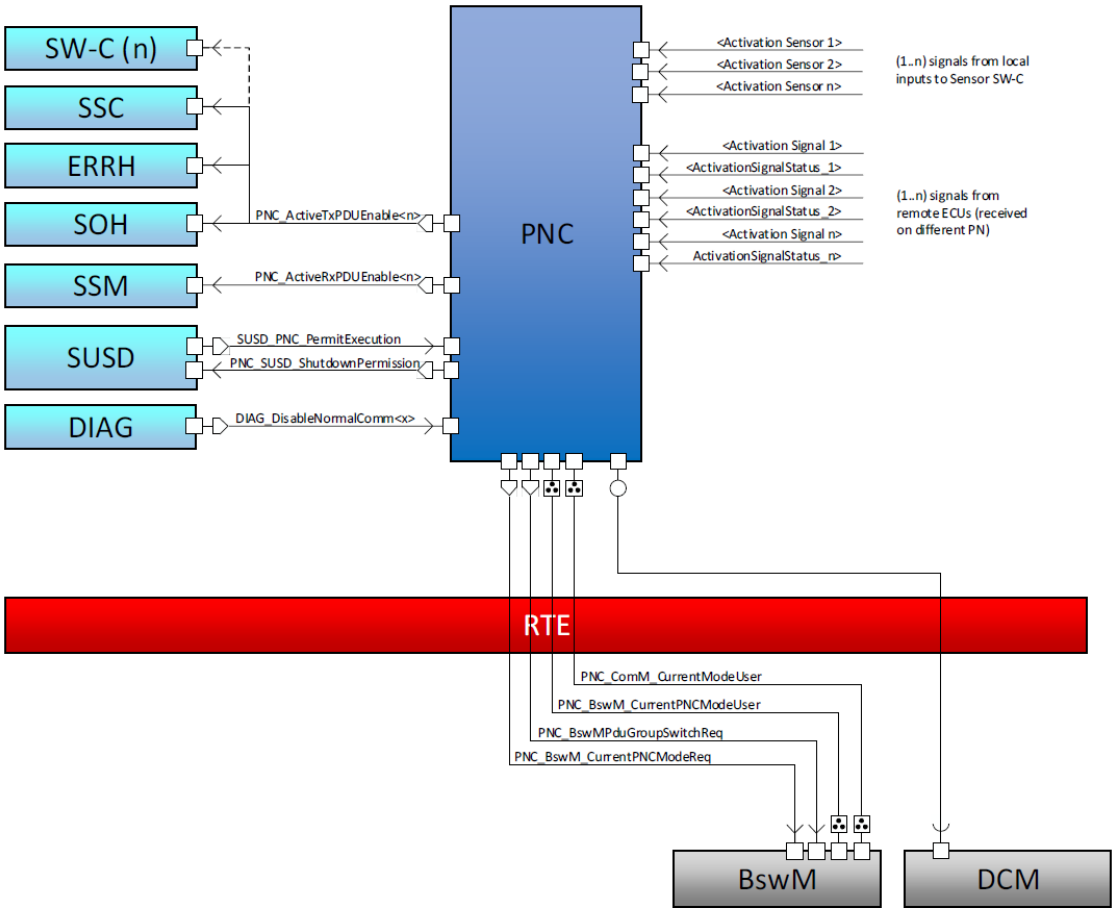


Figure 2.1: SUM_PNC Interfaces representation

Port Interface	Port Type	P-Port	R-Port	Description
PNC_ActiveTxPDUEnable<n>	S/R	SUM_PNC	SUM_SSC, SUM_ERRH and Local SW-Cs	Notification that PN<n> is transmitting signals
PNC_ActiveRxPDUEnable<n>	S/R	SUM_PNC	SUM_SSM	Notification to SSM that PN<n> is receiving signals
ShutdownPermission	S/R	SUM_PNC	SUM_SUSD	Notification to SUSD on activity/inactivity of PN
PermitExecution	S/R	SUM_SUSD	SUM_PNC	Interface to provide signal to enable start/stop of PNC
DIAG_DisableNormalComm<channel>	S/R	SUM_DIAG	SUM_PNC	Provides notification to SUM_PNC the status of diagnostics enabling or disabling of normal communications for each <channel>

Port Interface	Port Type	P-Port	R-Port	Description
<ActivationSignal_n>	S/R	SW-C (usually remote) or Dcm	SUM_PNC	n interface(s) for every serial data signal or RID that is involved in a PNs activation and/or deactivation criteria.
<ActivationSignal_Status_n>	S/R	SW-C (usually remote) or Dcm	SUM_PNC	n interface(s) for every serial data signal or RID that is involved in a PNs activation and/or deactivation criteria
<ActivationSensor_n>	C/S	Sensor SW-C	SUM_PNC	n C/S interface(s) to every sensor SW-C signal that is involved in a ECU's activation and/or deactivation criteria
PNC_BswM_CurrentPNCModeUser<n>	Mode User	BswM	SUM_PNC	Receives the status of PN<n> via BswM ComM_CurrentPNCMode for n PN(s) in which the ECU participates
PNC_BswM_CurrentPNCModeReq <n>	Mode Request	SUM_PNC	BswM	Request activation / deactivation of PN<n>. For Activating ECU only.
PNC_ComM_CurrentModeUser <n>	Mode User	BswM	SUM_PNC	Receives the status of the communication channels<n> of the ECU via BswM_ComMCurrentMode.
PNC_BswMPduGroupSwitchTxReq<n>	Mode Request	SUM_PNC	BswM	Request start / stop of I-PDU Tx group <n>
PNC_BswMPduGroupSwitchRxReq<n>	Mode Request	SUM_PNC	BswM	Request start / stop of I-PDU Rx group<n>
DataServices_PNC_DID_Info_Local	S/R	SUM_PNC	DCM	Interface to DCM to get the local partial network activation status (User Optional, if it is allowed by ECU type)
DataServices_PNC_DID_Info_Channel	S/R	SUM_PNC	DCM	Interface to DCM to get the communication channel status (User Optional, if it is allowed by ECU type)
DataServices_PNC_DID_Info_Remote	S/R	SUM_PNC	DCM	Interface to DCM on remote partial activation status. (User Optional, if it is allowed by ECU type)

Port Interface	Port Type	P-Port	R-Port	Description
PNC_ComMPNC_<n>_Group_DM_Switch	Mode Request	SUM_PNC	BswM	Request to enable or disable the deadline monitoring of the Rx PDU Group associated to PN<n>
PN_DTC_Allowed_<n>	S/R	SUM_PNC	SUM_SSM, SUM_SSC	Notification that a DTC against the PDUs of a specific PN can be set
SSM_C_Disable_<n>	S/R	SSM	PNC	This interface is used to read the C_Disable calibration published by SSM for Tx Pdu.
SUM_BswM_DisablePdu	Mode Switch	PNC	BswM	Interface with the BswM to disable the Pdu.

[$Satisfies $SRS_PNC 008] [$Satisfies $SRS_PNC 002] [$Satisfies $SRS_PNC 048] [$Satisfies $SRS_PNC 049] [$Satisfies $SRS_PNC 050] [$Satisfies $SRS_PNC 051] [$Satisfies $SRS_PNC 052] [$Satisfies $SRS_PNC 053] [$Satisfies $SRS_PNC 013] [$Satisfies $SRS_PNC 020] [$Satisfies $SRS_PNC 023] [$Satisfies $SRS_PNC 026]

NOTE:

1. The notation used for Naming the module: *SUM_<name of the module>*
2. SUM_PNC for Partial Network Coordinator;
3. This notation is used for naming interfaces: *<SUM Module Name>_<Interacting Module Name>_<Interface Name>*
   - Example: *PNC_BswM_CurrentPNCModeUser<n>*
4. This notation is used for representing states:
   [$Satisfies $SRS_<SUM Module Name> <requirement ID>]
5. This notation is used for specific APIs, where interactions are involved with BswM module: [RTE_callSpecificSuffix]_[RTE_Port_Name_<PartialNetworkID>_<ComMChannelID>_<PNCUserID>]_[VDP]]. Example:
   A. Rte_Mode_RP_BswM_ComM_Current_PNC_ModeUser0_6_5_PNC_State
   B. Rte_Write_PP_PNC_BswM_ComM_CurrentPNCMode_Req0_6_5_PNC_State

2.4 Assumptions and constraints

- All through the PN state machine, the condition check with naming convention **PN_Activation_Criteria<n>** is assumed to be a check for Local Activation and is maintained in that way all through the code;
- The user/integrator should ensure that the ComM Channels referred in "ComM_PNC_Ref" of Partial Network configuration under **PNC/PNCConfigSets/-SUM_PNC_PARTIAL_NETWORK_PARAMETERs** should match with ComM channels configured in "ComM_Channel_Ref" under **PNC/PNCGeneral/ComM_PNC_Channels**.

Design Decision

- All the Port interfaces will have ComMPnc ShortName (assumed to be following PN names specified in [R2]) appended in the end, e.g.:
  The RPort *RP_Bswm_ComM_Current_PNC_ModeUser_Propulsion* and the associated interface *PNC_BswM_ComM_CurrentPNCModeUser_Propulsion* are created for Propulsion Partial Network.

3 Building Block View Static View

3.1 Overview

[$SDD_PNC 034]
SUM_PNC manages all of the BSW services related to Partial Networks (PNs) for an ECU that is a member of any PN. SUM_PNC shall be implemented in all ECUs that are members of any PN. This allows all other SW-Cs to function abstracted from the source of the data (e.g. signals deployed to the serial communication networks).
Actions under taken in order to accomplish system goals, PNC shall,

i. Request to BswM to activate / deactivate PN per the activation / deactivation criteria.

ii. Publish notification that can be used by SW-Cs that participate in the PN, to know when the infrastructure services are ready to transmit and receive signals;

iii. Perform retries for PN activation / deactivation requests.

iv. SUM_PNC shall contain logic to compare the status of the Mode Request port interface to BswM with the corresponding Mode User port interface. SUM_PNC needs to make sure that the BSW state matches the requested state. SUM_PNC logic requests for PN activation needs to be in sync with the actual BSW PN activation status.

v. SW-C which sends signals via serial data communication shall use the appropriate return codes from the AUTOSAR communication / RTE layer to confirm transmission.

[$Satisfies $SRS_PNC 003] [$Satisfies $SRS_PNC 005] [$Satisfies $SRS_PNC 023] [$Satisfies $SRS_PNC 043]

The following diagram shows the main building blocks of the system. The interfaces and dependent SUMs, BSW modules are described in below.

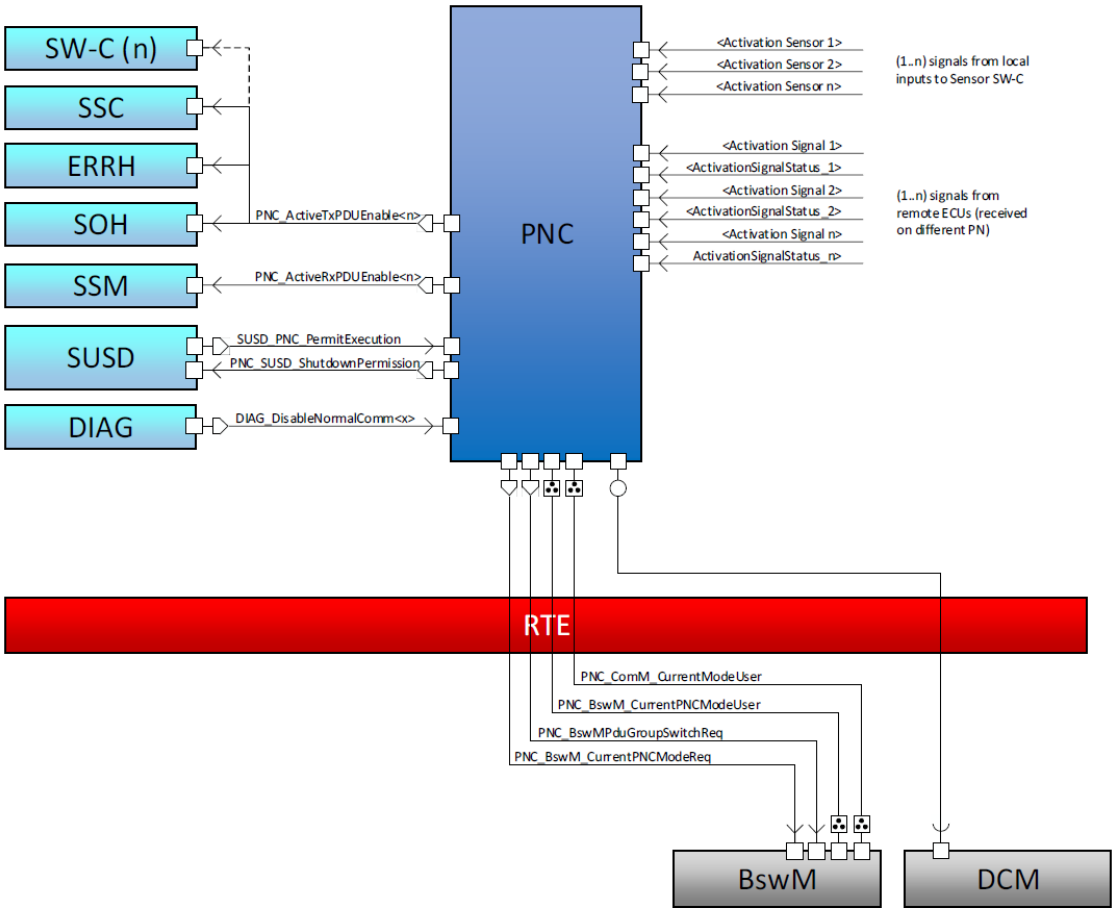


Figure 3.1: SUM_PNC Interfaces representation

3.1.1 SUM_SUSD

[$SDD_PNC 002]
Dependency and purpose of this module wrt PNC is mentioned below:

- The Standard Utility Module, Startup Shutdown (SUM_SUSD) contains the following functions: Application State Manager, Application SW-C Start /Shutdown, Operating/Battery Voltage Monitor, and Wakeup Reason DID.
- SUM_SUSD shall set the Wakeup(s) Active bit in the SUSD_PNC_PermitExecution interface to SUM_PNC to:
  - FALSE - when no wakeups relevant to PN activation are active
  - TRUE - when any wakeup(s) relevant to PN activation are active
- SUM_SUSD shall set the HW Ready bit in the SUSD_PNC_PermitExecution interface to SUM_PNC to:
  - FALSE - when Project Specific HW needed for PN activation is not ready
  - TRUE - when Project Specific HW needed for PN activation is ready
- PNC shall publish Signal PNC_ShutdownPermission to SUM_SUSD to inform if any PN is currently active or not.

[$Satisfies $SRS_PNC 013] [$Satisfies $SRS_PNC 008] [$Satisfies $SRS_PNC 010] [$Satisfies $SRS_PNC 011]

3.1.2 SUM_DIAG

[$SDD_PNC 003]
Dependency and purpose of this module wrt PNC is mentioned below:

- The Standard Utility Module, Diagnostic Application Service (SUM_DIAG), Preconditions to support diagnostic ECU Reset service, Preconditions to support Communication Control service request etc.
- SUM_DIAG provides notification about the status of diagnostics enabling or disabling of normal communications for each <channel> to PNC.
- SUM_PNC shall check disable normal communication from the tester to inhibit remote activation of a PN if not yet active as Passive ECUs do not need to remotely activate a PN when normal communication is disabled.

[$Satisfies $SRS_PNC 025]

3.1.3 SUM_SSM

[$SDD_PNC 035]
Dependency and purpose of this module wrt PNC is mentioned below:

- The Standard Utility Module, Signal Status Monitor (SUM_SSM)’s scope is for handling status information related to the Signal Integrity and Source Integrity for received signals, and for handling of generating/checking safety signal transmitted/received with in a protected signal group.
- SUM_PNC shall publish Signal PNC_ActiveRxPDUEnable<n>_Enable to SUM_SSM to notify PN<n> rx status (enabled / disabled).
- SUM_SSM shall transition to SIG_STATUS_DISABLED by use of the appropriate SUM_PNC notifica-

tion, PNC_SWC_PN<n>Enable, to be notified when all partial networks for the associated signal are inactive [$Satisfies $SRS_PNC 026]

- SUM_PNC shall publish PN_DTC_Allowed<n> to SUM_SSM so that SUM_SSM updates the PN_DTC_Enabled Flags related to Loss of Comm or Invalid Data DTCs.

3.1.4 SUM_SOH

Dependency and purpose of this module wrt PNC is mentioned below:

- The Standard Utility Module, State of Health manager (SUM_SOH) contains the following functions: Aggregation/Monitoring of SW platform modes (BSW States), WdgM SE states, reset requests and their occurrences.
- SUM_PNC shall publish Signal PNC_ActiveTxPDUEnable<n> to SUM_SOH, that PN<n> failed attempt to start
- SUM_SOH shall only publish the reliability monitor events signal when the PNC_ActiveTxPDUEnable interface is TRUE. If the PNC_ActiveTxPDUEnable interface is FALSE, then SOH shall retain the reliability monitor events signal in its output buffer.

3.1.5 SUM_ERRH

Dependency and purpose of this module wrt PNC is mentioned below:

- The Standard Utility Module, Error Handler (SUM_ERRH), is notified of error events created by BSW Production Errors or SUM error events;
- SUM_ERRH shall implement a port, PNC_ActiveTxPDUEnable, to receive notification of the availability of the Partial Network in which the message containing the DTC Triggered signal group participates;
- SUM_ERRH shall only publish the DTC Triggered signal when PNC_ActiveTxPDUEnable is TRUE. If PNC_ActiveTxPDUEnable is FALSE, then SUM_ERRH shall retain the DTC Triggered message in its output buffer;
- SUM_PNC shall publish Signal PNC_ActiveTxPDUEnable<n> to SUM_SSC, SUM_SOH, and SUM_ERRH. Publish notification to notify PN<n> tx status (enabled / disabled).

3.1.6 SUM_SSC

[$SDD_PNC 006]
Dependency and purpose of this module wrt PNC is mentioned below:

- The Standard Utility Module, Security Service Coordinator (SUM_SSC), is responsible for functions like Transmitter of authenticated messages, transmitter of non-authenticated messages having ARC, PV, CS/CSR etc.
- SUM_PNC shall publish Signal PNC_ActiveTxPDUEnable<n> to SUM_SSC, to notify PN<n> tx status (enabled / disabled).
  [$Satisfies $SRS_PNC 026]
- SUM_PNC shall publish PN_DTC_Allowed<n> to SUM_SSC so that SUM_SSC updates the PN_DTC_Enabled Flags related to the U1962 DTC.

3.1.7 BSW Elements: System, Services

[$SDD_PNC 043]

BswM

- AUTOSAR Basic Software module BSW Mode Manager (BswM), Its responsibility is to arbitrate mode requests from application layer SW-Cs or other BSW modules based on simple rules, and perform actions based on the arbitration result.
- SUM PNC shall have mode requester interface and mode switch interface to get information on Current PNC mode, request to enable/disable relevant Tx/Rx groups and PDUs.

Mode Requester interface (PNC-BSWM)	**Description**	**Info obtained via**
PNC_BswM_CurrentPNCModeReq	PNC_BswM_CurrentPNCModeReq to request the appropriate PN(s) to be activated / deactivated	ComM
PNC_BswMPduGroupSwitchTxReq	To request enabling / disabling relevant Tx PDU groups	Com
PNC_BswMPduGroupSwitchRxReq	To request enabling / disabling relevant Rx PDU groups	Com
PNC_BswM_PDUGroup_DM_Switch	To request enabling / disabling of Rx PDU Group deadline monitoring	Com
SUM_BswM_DisablePdu	To request disabling of Tx PDUs according to SSM indication.	SSM

Mode User interface(PNC-BSWM)	**Description**	**Info obtained via**
PNC_BswM_CurrentPNCModeUser	To inform the current state of the PN	ComM
PNC_ComM_CurrentModeUser	To inform the current state of channels related to the PN(s) for this ECU	ComM

- Requestor of Partial Network will use following PN states for Mode Requests for function PNC_BswM_CurrentPNCModeReq <n>
  a. SUM_PNC_NO_COMMUNICATION
  b. SUM_PNC_REQUESTED
- Mode Switch Interface PNC_BswM_CurrentPNCModeUser<n> for individual PN will return the current PN state using all states defined below:
  a. SUM_PNC_NO_COMMUNICATION
  b. SUM_PNC_PREPARE_SLEEP
  c. SUM_PNC_READY_SLEEP
  d. SUM_PNC_REQUESTED

- Mode switch interface PNC_ComM_CurrentModeUser <n> to indicate current state of ComM for the desired communication channel. The ComM Channel states are defined as below:

  a. SUM_COMM_NO_COMMUNICATION
  b. SUM_COMM_FULL_COM_READY_SLEEP
  c. SUM_COMM_FULL_COM_NETWORK_REQUESTED
  d. SUM_COMM_FULL_COMMUNICATION
  e. SUM_COMM_SILENT_COMMUNICATION

- Mode Request Interface PNC_BswMPduGroupSwitchTxReq<n> and PNC_BswMPduGroupSwitchRxReq<n> (to Com) to request for Tx or Rx PDU Group enabling associated with specific Partial networks.

  a. SUM_PDUGRP_STOP
  b. SUM_PDUGRP_START

  a. SUM_COMM_NO_COMMUNICATION
  b. SUM_COMM_FULL_COM_READY_SLEEP
  c. SUM_COMM_FULL_COM_NETWORK_REQUESTED
  d. SUM_COMM_FULL_COMMUNICATION
  e. SUM_COMM_SILENT_COMMUNICATION

- Mode Request Interface SUM_BswM_DisablePdu to request disabling of specific Tx pdus according to what SUM_SSM communicates through the SSM_C_Disable_Tx port interface.

  a. PDU_DISABLE_REQ
  b. PDU_ENABLE_REQ

- Mode Request Interface PNC_BswM_PDUGroup_DM_Switch<n> to request enabling or disabling the deadline monitoring for the Rx PDU group of the selected PN:

  a. PN_DM_ENABLE
  b. PN_DM_DISABLE

ComM

1. The ComM module collects the bus communication access requests from communication requestors and coordinates the bus communication access requests.
2. ComM updates SUM PNC via BswM the Current mode and current PNC mode when requested.

Com

- AUTOSAR Com module provide signal gateway functionality, routing of signals/signal groups from received I-Pdus to be transmitted, communication transmission control (start/stop of I-pdu groups) etc functionalities.
- Com shall start/stop the IPdu group with help of its service Com_IpduGroupControl. When PNC_BswMPduGroupSwitchTxReq<n > and PNC_BswMPduGroupSwitchRxReq<n> is requested via BswM, this api sets the activation state of all I-Pdu groups to requested state first and then enables/disables the group.

CanSM

- CanSM module shall referenced in PNC to get the Repeat Message Timer value.

[$Satisfies $SRS_PNC 003] [$Satisfies $SRS_PNC 014] [$Satisfies $SRS_PNC 015] [$Satisfies $SRS_PNC 018] [$Satisfies $SRS_PNC 022] [$Satisfies $SRS_PNC 023]

3.2 Configuration

Module Name	**SUM_PNC**
Description	Configuration of SUM_PNC

Included Containers

Container Name	**Multiplicity**	**Scope/Description**
PNCConfigSet	1..1	This Container shall describe multiple configuration sets of PNC. This container and its sub container exist once per configuration set.
PNCGeneral	1..1	This container shall define all the general configuration parameter of PNC module.

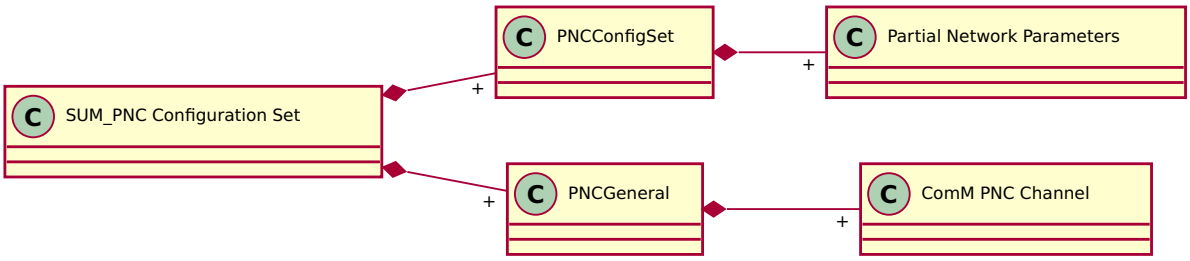


Diagram 3.1

3.2.1 General

This section shall describe the General Configuration parameters in detail

Container Name	**General**
Description	Specified the General Configuration for SUM_PNC

Container Name	Multiplicity	Scope
ComM_PNC_Channels	1..8	This container shall capture all the ComM channel references and CAN number configured for this System

[$SDD_PNC 004]

Parameter Name	SUM_PNC_ACTIVE_SYNCH_TIME
Description	Specifies Time in msec, for the PNC to request the PN Tx PDU group activation, after detecting that the activation criteria is met and after the activation of the Rx PDU group activation.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..100
Default value	60

[$Satisfies $SRS_PNC 055] [$Satisfies $SRS_PNC 032]

[$SDD_PNC 036]

Parameter Name	SUM_PNC_PASSIVE_SYNCH_TIME
Description	Specifies Time in msec, for the PNC to request the PN Tx PDU group activation, after detecting remote activation and after the activation of the Rx PDU group activation.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..100
Default value	10

[$Satisfies $SRS_PNC 055]

[$SDD_PNC 037]

Parameter Name	PNC_t_RemoteSignals_Periodicity
Description	Specifies Time in msec, for the periodicity of polling for changes to Activation Signals.
Multiplicity (upper..lower)	1..1
Category	ECUC-FLOAT-PARAM-DEF
Range	1.00 ..100.00
Default value	10.00

[$Satisfies $SRS_PNC 055]

[$SDD_PNC 038]

Parameter Name	PNC_t_PNsPNsStatusMonitoring_Periodicity
Description	Specifies Time in msec, for the periodicity of monitoring for changes to the PN status
Multiplicity (upper..lower)	1..1
Category	ECUC-FLOAT-PARAM-DEF
Range	1.00 ..100.00
Default value	10.00

[$Satisfies $SRS_PNC 055]

Parameter Name	Shutdown_permission_SUSD_Periodicity
Description	Specifies Time in msec, for the periodicity of Shut-down_permission runnable
Multiplicity (upper..lower)	1..1
Category	ECUC-FLOAT-PARAM-DEF
Range	1.00 ..100.00
Default value	10.00

[$SDD_PNC 039]

Parameter Name	PNC_t_Activation_Req_Timer
Description	Specifies Time in msec, for the periodicity of monitoring for changes to the PN status
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..100
Default value	10

[$Satisfies $SRS_PNC 055]

[$SDD_PNC 007]

Parameter Name	PNC_PermitExecution_Mask
Description	Mask that can be OR’d with the value supplied by PermitExecution to set any "don’t care" bits to ’1’
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..255
Default value	190

[$Satisfies $SRS_PNC 009] [$Satisfies $SRS_PNC 010] [$Satisfies $SRS_PNC 055] [$Satisfies $SRS_PNC 011]

Following are the parameters added additionally apart from ones specified in **[R1]**

[$SDD_PNC 101]

Parameter Name	PNC_t_MainFunction_Periodicity
Description	Specifies Time in msec, for the periodicity of polling for changes to PNC state changes.
Multiplicity (upper..lower)	1..1
Category	ECUC-FLOAT-PARAM-DEF
Range	1.00 .. 100.00
Default value	6.25

[$Satisfies $SRS_PNC 098]

Parameter Name	Number_Of_Retries
Description	Number of PNC Partial Networks
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..50
Default value	1

[$SDD_PNC 079]

Parameter Name	SUM_PN_Coordinator_Node_ID
Description	ECU ID (part of NM PDU) associated with CGM that is used for deadline monitoring control (used with SUM_PN_<n>_DM_Timer). Value in DECIMAL.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..255
Default value	69

[$Satisfies $SRS_PNC 055]

[$SDD_PNC 005]

Parameter Name	AR_GMSUM_4_2_X_SUPPORT
Description	This parameter is used to distinguish between the ComM V4.0.3 and V4.1.1and above. If the ComM V4.0.3 is used, ComMPncId will be configured in range of 0-47, for V4.1.1 and above, range is 8-63
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

[$Satisfies $SRS_PNC 060] [$Satisfies $SRS_PNC 061] [$Satisfies $SRS_PNC 062]

The following parameters are related to User Optional generation:

[$SDD_PNC 098]

Parameter Name	PncDidF08CCommunicationChannelStatusCanSupport
Description	This parameter shall be set by user in order to enable the generation of DID Info Channel port for Dcm
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	false

[$Satisfies $SRS_PNC 095]

[$SDD_PNC 099]

Parameter Name	PncGenerateInitRunnable
Description	It allows the generation of a periodic runnable to run function SUM_PNC_Init one time
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	false

[$Satisfies $SRS_PNC 097]

[$SDD_PNC 103]

Parameter Name	ComMPncIdLongRangeSupport
Description	This parameter is used to distinguish the range values that can be assigned to ComMPncId. If it is true the first comMPncId starts from 0, otherwise it starts from 16
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	true

[$Satisfies $SRS_PNC 099]

Parameter Name	SSMConfigSet_Ref
Description	This parameter is used to refer to the container SSMConfigSet of the SSM.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	NA

The following parameter is related to RTE version:

Parameter Name	RTEVersion5x
Description	Parameter to check if the RTE version 5.x to enable additional calls to access calibration parameter
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

ComM_PNC_Channels

Parameter Name	ComM_Channel_Ref
Description	List of all ComM channels which are configured for SUM PNC
Multiplicity (upper..lower)	1..*
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	NA

[$SDD_PNC 055]

Parameter Name	CAN_Channel_Number
Description	This parameter will configure the CAN number with associated ComM channel reference. This Parameter shall be used for updating Correct CAN channel status onto DID F08C (if it is present).
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..7
Default value	1

[$Satisfies $SRS_PNC 058]

3.2.2 Config Set

This section shall describe the Config set container configuration parameters in detail:

Container Name	PNCConfigSet
Description	Specifies the Configurations sets to configure multiple PNs for a given system

Included Container Name

Container Name	Multiplicity	Scope/Dependency
50	10	40
Container Name	**Multiplicity**	**Scope/Dependency**
SUM_PNC_PARTIAL_NETWORK_PARAMETERs	1..1	This container includes the parameters related to the Partial Network(s) of the system.

Partial Network Parameters

Parameter Name	IsRxEnabled
Description	Specifies if the RxPDU and RxPduGroup Interfaces are available for the PN or not. TRUE will create interfaces for for SSM and BswM.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

Parameter Name	IsTxEnabled
Description	Specifies if the TxPDU and TxPduGroup Interfaces are available for the PN or not. TRUE will create interfaces for for SSM and BswM.
Multiplicity (upper..lower)	1..1
Category	ECUC-BOOLEAN-PARAM-DEF
Range	NA
Default value	1

Parameter Name	Number_of_PNC_Activation_Sensors
Description	Specifies Number_of_PNC_Activation_Sensors per PN.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..10000
Default value	1

Parameter Name	Number_of_PNC_Activation_Signals
Description	Specifies Number_of_PNC_Activation_Signals per PN.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..10000
Default value	1

[$SDD_PNC 080]

Parameter Name	SUM_PN_DM_Timer
Description	Time in ms, for the PNC to request deactivation of the deadline monitoring of the Rx PDU group associated with the PN
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..8000
Default value	7000

[$Satisfies $SRS_PNC 055]

[$SDD_PNC 094]

Parameter Name	SUM_PN_DTC_TIMER
Description	Initial value of the calibration parameter SUM_PN<n> DM_Timer: time in ms to delay in setting of loss of comm, invalid data DTC and U1962.
Multiplicity (upper..lower)	1..1
Category	ECUC-INTEGER-PARAM-DEF
Range	0..5000
Default value	5000

[$Satisfies $SRS_PNC 056]

Parameter Name	ComM_PNC_Ref
Description	Reference to ComM PNC reference.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	NA

[$SDD_PNC 095]

Parameter Name	Repeat_Message_Time_Ref
Description	If CanNm is configured as PN Network Manager, this parameter is the reference for CanNmRepeatMessageTime parameter.Otherwise, if UdpNm is configured, this parameter is the reference for UdpNmRepeatMessageTime parameter. They provide the Timeout for Repeat Message State of the corresponding Network Manager.
Multiplicity (upper..lower)	1..1
Category	ECUC-REFERENCE-DEF
Range	NA
Default value	NA

[$Satisfies $SRS_PNC 037] [$Satisfies $SRS_PNC 041] [$Satisfies $SRS_PNC 094]

4 Runtime View Dynamic View

[$SDD_PNC 040]
The behavior of SUM PNC is described below as part of dynamic view:

1. SUM_PNC shall be initialized following the initialization of RTE, SchM and SUM_SUSD, and prior to the initialization of any other functional SW-Cs.
2. SUM_PNC shall implement a port for Permit Execution, which can receive notification from SUM_SUSD to execute its functions or prepare for shutdown.
3. SUM_PNC may use the configuration PNC_PermitExecution_Mask in its logic to allow configuration of which Permit Exectution bits are considered to trigger a start of its functions.
4. If SUM_PNC has no project specific dependency on HW, then SUM_PNC shall start execution of its functions when the Wakeup(s) Active bit in the Permit Execution interface is TRUE.
5. If SUM_PNC has project specific dependency on HW, then SUM_PNC shall execute its functions when both the HW Ready bit and the Wakeup(s) Active bit in the Permit Execution interface is TRUE.
6. When the ECU Reset is requested in the Permit Execution, then SUM_PNC shall set its Shutdown Permission to enable to support ECU reprogramming, even if there is one or more currently active Partial Networks.
7. SUM_PNC shall implement a port, Shutdown Permission, which can publish notification to SUM_SUSD to control the disable or enable shutting down the ECU.
8. Configuration parameters are assumed to be pre-compile, which means the ECU behaviour modification requires a reflash of the complete application software.
9. Calibration parameters are assumed to be post-build, which means the ECU behaviour can be modified by only reflashing the calibration software.

[$Satisfies $SRS_PNC 002] [$Satisfies $SRS_PNC 008] [$Satisfies $SRS_PNC 009] [$Satisfies $SRS_PNC 010] [$Satisfies $SRS_PNC 011] [$Satisfies $SRS_PNC 012] [$Satisfies $SRS_PNC 013]

4.1 System initialization

[$SDD_PNC 041]

- PNC_Init() API checks the activation criteria is FALSE and PNC_BswM_currentPNCModeUser is PNC_NO_COMM;
- If both the conditions are satisfied then it sets Current_PNC_Mode as PNC_Enable_Check (TRUE). Else it will:
  - i Initialize the activation criteria for local and remote ECUs as false
  - ii Initialize activation criteria of remote signal status
  - iii Initialize Current PNC State to SUM_PNC_NO_COMMUNICATION

iv Initialize ComM Channel State to SUM_COMM_NO_COMMUNICATION

v Initialize PduGroup_State to SUM_PDUGRP_STOP

vi Initialize Current_PNC_Mode to PNC_Enable_Check

vii Initialize the DM FSM to DM_ACTIVE state

viii Initialize DM timer to the configured value

ix Initialize PN_DTC_Timers to the configured value

x Intialize all PN_DTC_Allowed ports to DTC_Not_Allowed

- SUM_PNC_Mainfunction() API is called periodically at the rate specified with configuration parameter PNC_t_MainFunction_Periodicity.
- The Current_PNC_Mode is set to PNC_Enable_Check during initialization.
- The DID related interfaces are implemented as user optional and ComM channel specific check during transition from PNC_ACK_ACTIVATION to PNC_TX_ENABLE is implemented in API PNC_BswM_ComM_Current_ModeUSer().

NOTE: Super states IDLE and ACTIVE are not implemented as there is only one transition which can be handled in transition from Sub state PNC_REMOTLY_ACTIVE to sub state PNC_ENABLE_CHECK.
[$Satisfies $SRS_PNC 001] [$Satisfies $SRS_PNC 002] [$Satisfies $SRS_PNC 017] [$Satisfies $SRS_PNC 024]

4.2 System de-initialization

<N/A>

4.3 Wake-up procedure

[$SDD_PNC 045]

- Partial Network activators will be assigned criteria that describes the conditions, which the ECU will evaluate, to determine if a Partial Network should be activated or deactivated;
- Following Diagram describes the PN states and Sub-states within, these determine how SUM_PNC does Partial network activation/deactivation.

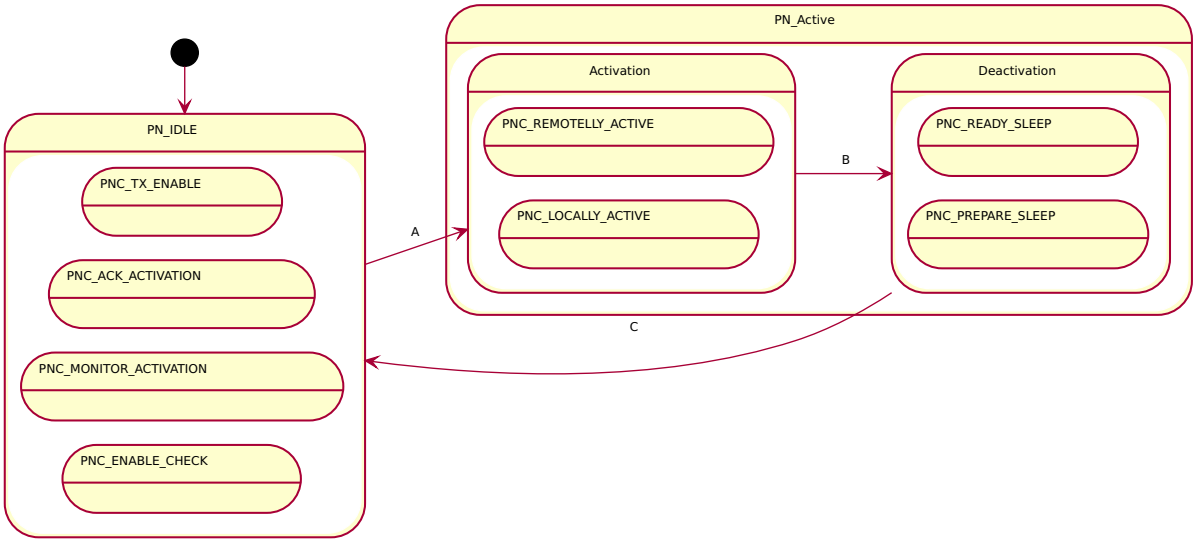


Diagram 4.1: Partial Network Activation and Deactivation

The Activation of a PN can be locally or via Remote activation. Both these methods are explained below.

[$Satisfies $SRS_PNC 021]

4.3.1 *Local Activation*

[$SDD_PNC 029]

Local activation is via SW-C local input signal. Local activation is explained in the next diagram:

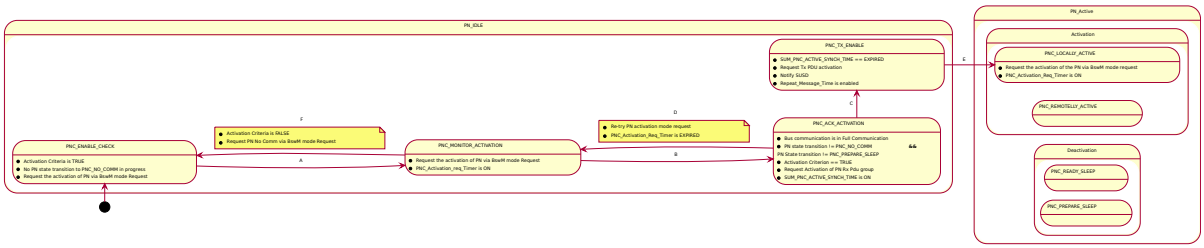


Diagram 4.2: PN Local Activation

[$Satisfies $SRS_PNC 007] [$Satisfies $SRS_PNC 021] [$Satisfies $SRS_PNC 031] [$Satisfies $SRS_PNC 032] [$Satisfies $SRS_PNC 033] [$Satisfies $SRS_PNC 034] [$Satisfies $SRS_PNC 036] [$Satisfies $SRS_PNC 037] [$Satisfies $SRS_PNC 102]

[$SDD_PNC 046]
When SUM_PNC determines that the local Activation criteria are false and Repeat Message Timer has expired or not started, Locally Active changes to Remotely Active state:

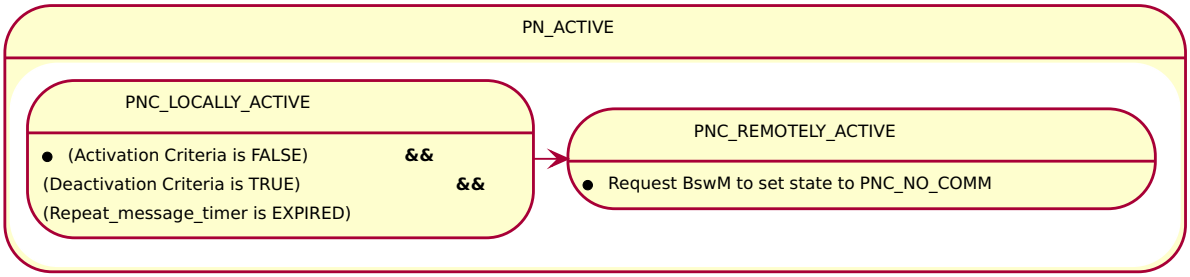


Diagram 4.3: PN Transition to Locally active to Remotely Active

[$Satisfies $SRS_PNC 041]

4.3.2 *Remote Activation*

[$SDD_PNC 030]

Remote activation is explained below:

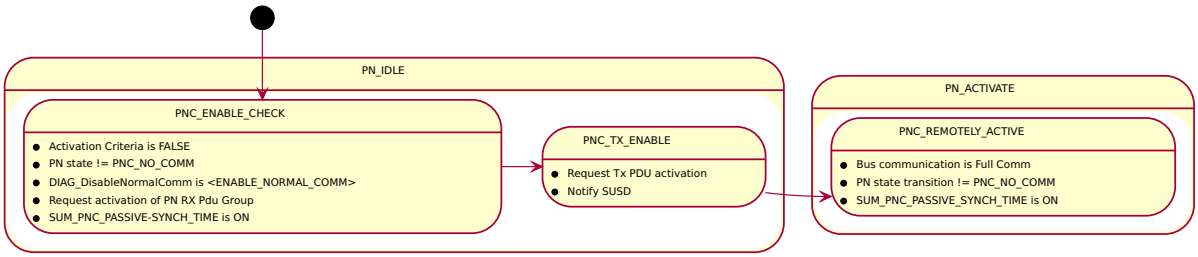


Diagram 4.4: Remote Activation

When SUM_PNC determines that Activation criteria for remote activation are false and local Activation criteria are true, following actions are executed:

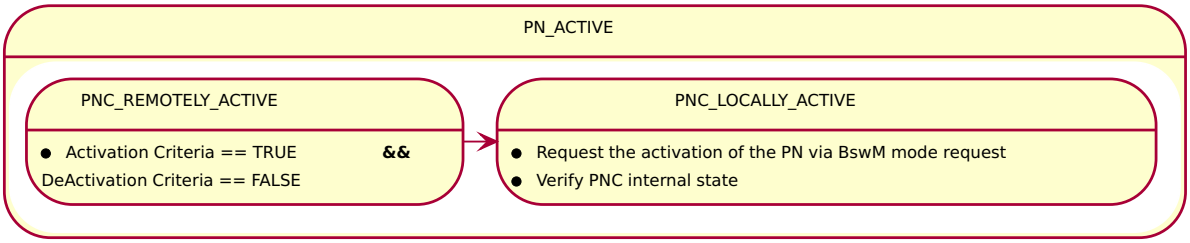


Diagram 4.5: Transition from rremotely to locally active

Afterwards, if SUM_PNC determines that Activation criteria for remote activation are true and local Activation Criteria are false and the Repeat Message Timer has not started or expired, SUM_PNC changes the PN state from Locally Active to Remotely Active.

[$Satisfies $SRS_PNC 007] [$Satisfies $SRS_PNC 021] [$Satisfies $SRS_PNC 038] [$Satisfies $SRS_PNC 039] [$Satisfies $SRS_PNC 040]

4.4 Sleep Procedure

[$SDD_PNC 042]

The sleep procedure is as described as below when SUM_PNC is notified of AUTOSAR BSW PN mode user state is PNC_PREPARE_SLEEP.

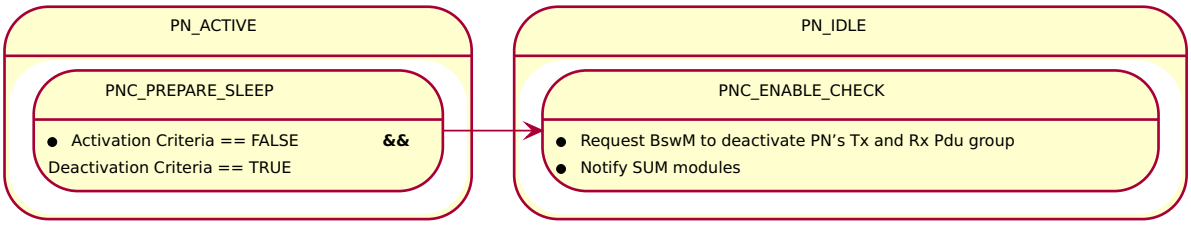


Diagram 4.6: Sleep Procedure

[$Satisfies $SRS_PNC 042] [$Satisfies $SRS_PNC 021]

4.5 Deadline Monitoring Timer

[$SDD_PNC 081]
SUM_PNC implements a timer SUM_PN<n>_DM_Timer and a FSM for each PN. The following states are implemented:

- DM_ACTIVE: default state. The deadline monitoring is enabled for that PN.
- DM_INACTIVE: deadline monitoring is disabled for that PN.

Deadline monitoring FSM is executed only when SUM_PNC is in PNC_REMOTELY_ACTIVE state. When SUM_PNC exits from PNC_REMOTELY_ACTIVE state, the DM_ACTIVE status is immediately set and a BswM request is sent to enable the deadline monitoring of the corresponding PN.
When SUM_PNC enters the PNC_REMOTELY_ACTIVE state, the deadline monitoring timer is periodically decremented. If the timer expires, the DM_INACTIVE state is entered, and a request is sent to BswM to disable the deadline monitoring of the corresponding PN. A transition from DM_INACTIVE to DM_ACTIVE state happens when the timer has been set to the configuration value, and a request is sent to BswM to enable the deadline monitoring of the corresponding PN.

SUM PNC implements a NM callback that is used to process all the received NM Frames. PN timers are set to the configuration value each time that the corresponding PN is remotely requested by the PN Coordinator ECU (ECU ID configurable through *PNCGeneral/SUM_PN_Coordinator_Node_ID*).

NOTE: In this implementation it is assumed that the NM callback is executed before a new NM frame is processed by CanNm.

[$Satisfies $SRS_PNC 064]

4.6 PN DTC Delay Timer

[$SDD_PNC 085]
SUM_PNC implements a decrementing timer *SUM_PN_DTC_TIMER<n>* for each configured PN.
[$Satisfies $SRS_PNC 080] [$Satisfies $SRS_PNC 081] [$Satisfies $SRS_PNC 082]

[$SDD_PNC 086]
The default value of this timer is read from *SUM_PN_DTC_TIMER<n>* calibration parameter.
[$Satisfies $SRS_PNC 056]

[$SDD_PNC 087]
SUM_PNC sets the timer to the default value on ECU start up and on deactivation of the associated PN.
[$Satisfies $SRS_PNC 080]
[$SDD_PNC 088]
SUM_PNC starts the timer for a PN when the PN state is either PNC_LOCALLY_ACTIVE or PNC_REMOTELY_ACTIVE.
[$Satisfies $SRS_PNC 081]]
[$SDD_PNC 089]
SUM_PNC stops decrementing the timer for a PN and resets it to the default value when the PN state exits from PNC_LOCALLY_ACTIVE or PNC_REMOTELY_ACTIVE states.
[$Satisfies $SRS_PNC 082]

[$SDD_PNC 090]
For each decrementing timer *SUM_PN_DTC_TIMER<n>* SUM_PNC creates and associates an output port of the S/R interface *PN_DTC_Allowed<n>*. The output parameter of such port interface can assume any of the following values:

- 0 = DTC not allowed for the particular PN;
- 1 = DTC allowed for the particular PN;
- 2 = DTC might be allowed for the particular PN

[$Satisfies $SRS_PNC 086] [$Satisfies $SRS_PNC 087]
[$SDD_PNC 091]
When the timer for a particular PN reaches zero, SUM_PNC shall set the *PN_DTC_Allowed<n>* port interface to 1 (DTC Allowed for this PN). [$Satisfies $SRS_PNC 083]
[$SDD_PNC 092]
When the value of the timer for a particular PN is equal to the default value, SUM_PNC shall set the *PN_DTC_Allowed<n>* port interface to 0 (DTC Not Allowed for this PN). [$Satisfies $SRS_PNC 084]
[$SDD_PNC 093]
When the timer for a particular PN is decrementing, SUM_PNC shall set the *PN_DTC_Allowed<n>* port interface to 2 (PN Timer Decrementing).
[$Satisfies $SRS_PNC 085]

4.7 Boot loading procedure

<N/A>

4.8 Failure handling strategy

- In case of inactivity of an PN, the interface to SUSD determines execution of SUM_PNC
- Also interface to other SUMs like SOH, ERRH is notified by SUM_PNC in case of inactivity or unexpected error.

4.9 Critical timing constraints

<N/A>

4.10 Tasks management

- Assumed that this Module is not a safety critical module as we are using EXPLICIT non-queued communications.

4.10.1 Periodic Runnable

Periodic runnables can be of two types:

1. Fixed periodicity that cannot be configured;
2. Fixed peridocity that can be configured via configuration parameter;

[$SDD_PNC 102]
For the second ones their periodicity can be expressed as a real number and it shall be considered as a time value in microseconds.
[$Satisfies $SRS_PNC 098]

SUM_PNC_MainFunction

[$SDD_PNC 008]

Name	SUM_PNC_Mainfunction
Periodicity	Configured via PNC_t_MainFunction_Periodicity
Inter-runnable Variables	NA
Managed Input Interfaces	SUSD_PNC_PermitExecution DIAG_DisableNormalComm ActivationSensor<n> ActivationSignal<n> ActivationSignalStatus<n>
Managed Output Interfaces	PNC_ActiveTxPDUEnable<n> PNC_ActiveRxPDUEnable<n> PNC_BswM_CurrentPNCModeReq PNC_BswMPduGroupSwitchReq PN_DTC_Allowed<n>
Per Instance Memory	NA
Return	void
Description	Manages status of all the PNs which ECU is part of

[$Satisfies $SRS_PNC 022] [$Satisfies $SRS_PNC005] [$Satisfies $SRS_PNC 006] [$Satisfies $SRS_PNC 016] [$Satisfies $SRS_PNC 017] [$Satisfies $SRS_PNC 024] [$Satisfies $SRS_PNC 025] [$Satisfies $SRS_PNC 026] [$Satisfies $SRS_PNC 030] [$Satisfies $SRS_PNC 033] [$Satisfies $SRS_PNC 034] [$Satisfies $SRS_PNC 042] [$Satisfies $SRS_PNC 043] [$Satisfies $SRS_PNC 019] [$Satisfies $SRS_PNC 044] [$Satisfies $SRS_PNC 056]

Runnable_Pnc_Init

[$SDD_PNC 100]

Name	Runnable_Pnc_Init
Periodicity	1000ms
Inter-runnable Variables	NA
Managed Input Interfaces	NA
Managed Output Interfaces	PN_DTC_Allowed
Per Instance Memory	NA
Return	void
Description	Executes SUM_PNC_Init() just once as a periodic runnable. this one is present **if and only if** the configuration parameter *GeneratePncInitRunnable* is TRUE, otherwise the user shall call SUM_PNC_Init() in the BswM initialization action

[$Satisfies $SRS_PNC 097]

Activation_Signal_In

[$SDD_PNC 009]

Name	Activation_Signal_In
Periodicity	Configured via PNC_t_RemoteSignals_Periodicity
Inter-runnable Variables	NA
Managed Input Interfaces	ActivationSignal<n>
Managed Output Interfaces	NA
Per Instance Memory	NA
Return	void
Description	This runnable reads activation criteria from remote ECUs and helps in setting the PNC mode

[$Satisfies $SRS_PNC 020]

Activation_Signal_Status_In

[$SDD_PNC 010]

Name	Activation_Signal_Status_In
Periodicity	Configured via PNC_t_PNsPNsStatusMonitoring_Periodicity
Inter-runnable Variables	NA
Managed Input Interfaces	ActivationSignalStatus<n>
Managed Output Interfaces	NA
Per Instance Memory	NA
Return	void
Description	This runnable reads activation criteria from remote ECUs and helps in setting the PNC mode

[$Satisfies $SRS_PNC 020]

Activation_Sensor

[$SDD_PNC 011]

Name	`Activation_Sensor`	
Periodicity	50ms	
Inter-runnable Variables	*in*	Activation_Criteria_out
Managed Input Interfaces	ActivationSensor<n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	void	
Description	This runnable provides the activation criteria from Signals that are read by a Sensor SW-C, local to the ECU.	

[$Satisfies $SRS_PNC 020]

Shutdown_Permission_SUSD

[$SDD_PNC 012]

Name	`Shutdown_permission_SUSD`
Periodicity	50ms
Inter-runnable Variables	NA
Managed Input Interfaces	NA
Managed Output Interfaces	PNC_SUSD_ShutdownPermission
Per Instance Memory	NA
Return	void
Description	This runnable manages the ShutdownPermission interface port connected to SUM_SUSD. If at least one PN is active, then SUM_PNC sets the status of ShutdownPermission to disabled (0 to 3). If no PNs are active, then SUM_PNC sets the status of ShutdownPermisison to enabled (0xFF).

[$Satisfies $SRS_PNC 013] [$Satisfies $SRS_PNC 063]

PNC_BswM_ComM_Pdugroupswitch_RxReq

[$SDD_PNC 014]

Name	`PNC_BswM_ComM_Pdugroupswitch_RxReq`	
Periodicity	50ms	
Inter-runnable Variables	*in*	Arg_PduGroup_State
Managed Input Interfaces	PNC_BswMPduGroupSwitchReq	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	Std_ReturnType	
Description	This runnable Writes PduGroupRX_State to SUM_SSM	

[$Satisfies $SRS_PNC 014] [$Satisfies $ SRS_PNC 022]

PNC_Active_TxPDU_Enable

[$SDD_PNC 015]

Name	PNC_Active_TxPDU_Enable	
Periodicity	50ms	
Inter-runnable Variables	*in*	ATxPDUEnable
Managed Input Interfaces	PNC_ActiveTxPduEnable<n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	Std_ReturnType	
Description	This runnable Sends Notification from SUM_PNC that Tx PDU groups for PN<n> are transmitting signals.	

[$Satisfies $SRS_PNC 026]

PNC_Active_RxPDU_Enable

[$SDD_PNC 016]

Name	PNC_Active_RxPDU_Enable	
Periodicity	50ms	
Inter-runnable Variables	*in*	ARxPDUEnable
Managed Input Interfaces	PNC_ActiveRxPduEnable<n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	Std_ReturnType	
Description	This runnable Sends Notification from SUM_PNC that Rx PDU groups for PN<n> are started/stopped.	

[$Satisfies $SRS_PNC 036] [$Satisfies $SRS_PNC 038]

PNC_BswM_ComM_Current_PNC_ModeUSer

[$SDD_PNC 017]

Name	PNC_BswM_ComM_Current_PNC_ModeUSer	
Periodicity	Mode Switch Event	
Inter-runnable Variables	*in*	NA
Managed Input Interfaces	PNC_BswM_CurrentPNCModeUser<n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	PNC_State	
Description	The port type is Mode user where bswM is P port and SUM_PNC is R Port. This runnable receives the status of the current PNC mode from Bswm.	

[$Satisfies $SRS_PNC 023] [$Satisfies $SRS_PNC 018]

PNC_ComM_Current_ModeUSer

[$SDD_PNC 018]

Name	PNC_ComM_Current_ModeUSer	
Periodicity	Mode Switch Event	
Inter-runnable Variables	*in*	NA
Managed Input Interfaces	PNC_ComM_CurrentModeUser <n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	ComMChannel_State	
Description	The port type is Mode user where bswM is P port and SUM_PNC is R Port. This runnable receives the status of the current communication channels<n> of the ECu via BswM_ComMCurrentMode.	

[$Satisfies $SRS_PNC 023]

PNC_BswM_ComM_CurrentPNCMode_Req

[$SDD_PNC 019]

Name	PNC_BswM_Current_PNC_Mode_Req	
Periodicity	Mode Switch Event	
Inter-runnable Variables	*in*	Arg_Current_PNC_State
Managed Input Interfaces	PNC_BswM_CurrentPNCModeReq <n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	Std_ReturnType	
Description	The port type is Mode requester where SUM_PNC is P port and BswM is R Port. This runnable request for activation/ deactivation of PN<n>	

[$Satisfies $SRS_PNC 022] [$Satisfies $SRS_PNC 018]

4.10.2 Event-Triggered Runnable

PNC_DataServicesPN_DID_Info_Channel

[$SDD_PNC 056]

Name	PNC_DataServicesPN_DID_Info_Channel	
Trigger	Operation invoked event	
Inter-runnable Variables	*in/out* *in/out*	OpStatus Data
Managed Input Interfaces	DataServices<PNC>Data<n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	Std_ReturnType	
Description	The port type is Client/Server where SUM_PNC is P port and DcM is R Port. This runnable requests DID for getting information on ECU operating state. This runnable is Optional: if ECU type does not prevent the DIDs, runnable shall be selected by User in order to be defined	

[$Satisfies $SRS_PNC 058] [$Satisfies $SRS_PNC 054]

PNC_DataServicesPN_DID_Info_Channel_CheckRead

[$SDD_PNC 057]

Name	PNC_DataServicesPN_DID_Info_Channel_CheckRead	
Trigger	Operation invoked event	
Inter-runnable Variables	*in/out* *in/out*	OpStatus ErrorCode
Managed Input Interfaces	DataServices<PNC>Data<n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	Std_ReturnType	
Description	The port type is Client/Server where SUM_PNC is P port and DcM is R Port. This runnable requests DID for getting information on Condition Check Read of ECU Operating State. This runnable is Optional: if ECU type does not prevent the DIDs, runnable shall be selected by User in order to be defined	

[$Satisfies $SRS_PNC 058]

PNC_DataServicesPN_DID_Info_Local

[$SDD_PNC 058]

Name	PNC_DataServicesPN_DID_Info_Local_CheckRead
Trigger	Operation invoked event
Inter-runnable Variables	*in* | OpStatus *in/out* | Data
Managed Input Interfaces	DataServices<PNC>Data<n>
Managed Output Interfaces	NA
Per Instance Memory	NA
Return	Std_ReturnType
Description	The port type is Client/Server where SUM_PNC is P port and DcM is R Port. This runnable requests DID getting information on ECU operating state in case of Local activation.

[$Satisfies $SRS_PNC 057] [$Satisfies $SRS_PNC 054]

PNC_DataServicesPN_DID_Info_Local_CheckRead

[$SDD_PNC 059]

Name	PNC_DataServicesPN_DID_Info_Local_CheckRead
Trigger	Operation invoked event
Inter-runnable Variables	*in/out* | OpStatus *in/out* | ErrorCode
Managed Input Interfaces	DataServices<PNC>Data<n>
Managed Output Interfaces	NA
Per Instance Memory	NA
Return	Std_ReturnType
Description	The port type is Client/Server where SUM_PNC is P port and DcM is R Port. This runnable requests DID for getting information on Condition Check Read of ECU Operating State in case of Local activation.

[$Satisfies $SRS_PNC 057]

PNC_DataServicesPN_DID_Info_Remote

[$SDD_PNC 060]

Name	PNC_DataServicesPN_DID_Info_Remote	
Trigger	Operation invoked event	
Inter-runnable Variables	*in* *in/out*	OpStatus Data
Managed Input Interfaces	DataServices<PNC>Data<n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	Std_ReturnType	
Description	The port type is Client/Server where SUM_PNC is P port and DcM is R Port. This runnable requests DID for getting information of ECU Operating State in case of remote activation.	

[$Satisfies $SRS_PNC 059] [$Satisfies $SRS_PNC 054]

PNC_DataServicesPN_DID_Info_Remote_CheckRead

[$SDD_PNC 061]

Name	PNC_DataServicesPN_DID_Info_Remote_CheckRead	
Trigger	Operation invoked event	
Inter-runnable Variables	*in/out* *in/out*	OpStatus ErrorCode
Managed Input Interfaces	DataServices<PNC>Data<n>	
Managed Output Interfaces	NA	
Per Instance Memory	NA	
Return	Std_ReturnType	
Description	The port type is Client/Server where SUM_PNC is P port and DcM is R Port. This runnable requests DID for getting information on Condition Check Read of ECU Operating State in case of remote activation.	

[$Satisfies $SRS_PNC 059]

4.10.3 Private Functions

SUM_PNC_Init

[$SDD_PNC 020]

Name	Sum_PNC_Init
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	void
Description	Initialization function for PNC module

[$Satisfies $SRS_PNC 017] [$Satisfies $SRS_PNC 001]

[$SDD_PNC 070]

All SUM_PN<n>_DM_Timers are set to default value – configured with SUM_PN_DM_Timer parameter – as part of SUM_PNC Initialization sequence.
[$Satisfies $SRS_PNC 065]

Set_Alarm

Name	`Set_Alarm`	
In parameters	`uint8` `uint8`	Alarm_Value PNC_Number
Out parameters	NA	
InOut parameters	NA	
Return	void	
Description	Handles the timer functionalities	

Permit_Execution

[$SDD_PNC 021]

Name	`Permit_Execution_SUSD`
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	boolean
Description	8 bit data received from SUM_SUSD, the wakeup bit of this value is compared with configured PNC_PermitExecution_Mask, if the wakeup bit is set, this function returns the permit execution permission as enabled else permit execution permission denied.

[$Satisfies $SRS_PNC 008] [$Satisfies $SRS_PNC 010] [$Satisfies $SRS_PNC 011] [$Satisfies $SRS_PNC 009]

Get_DIAG_DisableNormalComM

[$SDD_PNC 022]

Name	`Get_DIAG_DisableNormalCommunicationMode`
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	boolean
Description	This function receives the mode status from SUM_DIAG. If the Disable normal communication is ON, SUM_PNC will enable normal communication for each channel. If the Disable Normal communication is OFF, SUM_PNC will disable normal communication for each channel.

[$Satisfies $SRS_PNC 025]

PNC_Update_Local_Remote_DID_Bits

Name	PNC_Update_Local_Remote_DID_Bits
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	NA
Description	This Function is for setting the DID bits to indicate remote activation/Local activation of PNs in NM PU frame

Status_PNC_Enable_Check

[$SDD_PNC 023]

Name	Status_PNC_Enable_Check
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	void
Description	This Function is for handling PNC_ENABLE_CHECK state of PNC State chart

[$Satisfies $SRS_PNC 038]

Status_PNC_Monitor_Activation

[$SDD_PNC 024]

Name	Status_PNC_Monitor_Activation
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	void
Description	This Function is for handling PNC_MONITOR_ACTIVATION state of PNC State chart

[$Satisfies $SRS_PNC 031] [$Satisfies $SRS_PNC 032]

Status_PNC_Ack_Activation

[$SDD_PNC 025]

Name	Status_PNC_Ack_Activation
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	void
Description	This Function is for handling PNC_ACK_ACTIVATION state of PNC State chart

[$Satisfies $SRS_PNC 032] [$Satisfies $SRS_PNC 033] [$Satisfies $SRS_PNC 034]

Status_PNC_Tx_Enable

[$SDD_PNC 026]

Name	`Status_PNC_Tx_Enable`
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	void
Description	This Function is for handling PNC_TX_ENABLE state of PNC State chart

[$Satisfies $SRS_PNC 035] [$Satisfies $SRS_PNC 036] [$Satisfies $SRS_PNC 038] [$Satisfies $SRS_PNC 100] [$Satisfies $SRS_PNC 101]

Status_PNC_Locally_Active

[$SDD_PNC 027]

Name	`Status_PNC_Locally_Active`
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	void
Description	This Function is for handling PNC_LOCALLY_ACTIVE state of PNC State chart

[$Satisfies $SRS_PNC 037] [$Satisfies $SRS_PNC 041]

Status_PNC_Remotely_Active

[$SDD_PNC 028]

Name	`Status_PNC_Remotely_Active`
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	void
Description	This Function is for handling PNC_REMOTEY_ACTIVE state of PNC State chart

[$Satisfies $SRS_PNC 039] [$Satisfies $SRS_PNC 040]

[$SDD_PNC 071]
SUM_PN<n>_DM_Timer is disabled, i.e. the timer state is set to PNC_DM_ACTIVE, when the PN exits PNC_REMOTELY_ACTIVE state.
[$Satisfies $SRS_PNC 066]

[$SDD_PNC 084]
SUM_PNC invokes COM_ReceptionDMControl API on exit from PNC_REMOTELY_ACTIVE sub-state to enable deadline monitoring for the associated Rx PDU groups for PN<n> due to local activation of the PN.
[$Satisfies $SRS_PNC 079] [$Satisfies $SRS_PNC 064]

PNC_NM_Frame_Indication

Name	PNC_NM_Frame_Indication	
In parameters	NetworkHandleType	NetworkHandle
Out parameters	NA	
InOut parameters	NA	
Return	void	
Description	This function implements the NM callback needed for deadline monitoring. Sets the timer of requested PN by the PN Coordinator ECU	

[$SDD_PNC 073]
SUM_PNC implements the callback PNC_NM_Frame_Indication to be called by the Nm module when an NM frame is received by the ECU. This callback allows to determine the Partial Network status for all PNs supported by the individual ECU.
[$Satisfies $SRS_PNC 068] [$Satisfies $SRS_PNC 074]

[$SDD_PNC 074]
In the context of the callback PNC_NM_Frame_Indication, SUM_PNC evaluates the ECUID associated with SUM_PN_Coordinator_Node ID.
[$Satisfies $SRS_PNC 069] [$Satisfies $SRS_PNC 075]

[$SDD_PNC 075]
SUM_PNC evaluates individual partial network Identifier Bits (Bytes 2-7) extracted through CanNm_GetPduData API to determine if individual PNs relevant for the ECU is requested to be active. For ECUs supporting Partial Networking on Ethernet bus, SUM_PNC shall use UdpNm_GetPduData interface from UdpNm for obtaining the status information associated with each individual partial network.
[$Satisfies $SRS_PNC 070] [$Satisfies $SRS_PNC 076] [$Satisfies $SRS_PNC 098]

[$SDD_PNC 076]
Every time an individual partial network Identifier Bits (Bytes 2-7) indicates PN bit to be active (remotely active PN) SUM_PNC resets the SUM_PN<n>_DM_Timer to default value for PN<n> (configured with SUM_PN_DM_Timer parameter).
[$Satisfies $SRS_PNC 071]

PNC_DeadlineMonitoring

Name	PNC_DeadlineMonitoring	
In parameters	uint8	PN_index
Out parameters	NA	
InOut parameters	NA	
Return	void	
Description	Function that implements the deadline monitoring FSM.	

[$SDD_PNC 072]
SUM_PN<n>_DM_Timer is enabled when the PN enters the PNC_DM_ACTIVE state.
[$Satisfies $SRS_PNC 067]

[$SDD_PNC 077]

SUM_PNC transitions from PNC_DM_ACTIVE sub state to PNC_DM_INACTIVE sub state on expiry of the SUM_PN<n>_DM_Timer for PN<n>.
[$Satisfies $SRS_PNC 072]

[$SDD_PNC 078]
SUM_PNC invokes COM_ReceptionDMControl on entry into PNC_DM_INACTIVE sub state mode request to disable deadline monitoring for the associated Rx PDU groups for PN<n>.
[$Satisfies $SRS_PNC 073]

[$SDD_PNC 082]
SUM_PNC transitions from PNC_DM_INACTIVE sub state to PNC_DM_ACTIVE sub state on determination that an individual partial network Identifier Bits (Bytes 2-7) PN bit transitions from inactive to active.
[$Satisfies $SRS_PNC 077]

[$SDD_PNC 083]
SUM_PNC invokes COM_ReceptionDMControl on exit from PNC_DM_INACTIVE sub state mode request to enable deadline monitoring for the associated Rx PDU groups for PN<n>
[$Satisfies $SRS_PNC 078]

Check_DTC_Allowed_Timer

Name	`Check_DTC_Allowed_Timer`
In parameters	NA
Out parameters	NA
InOut parameters	NA
Return	NA
Description	Function that implements the PNC DTC Timer FSM.

PNC_BswM_ComM_Pdugroupswitch_TxReq

[$SDD_PNC 013]

Name	`PNC_BswM_ComM_Pdugroupswitch_TxReq`
In parameters	`PduGroup_State`
Out parameters	NA
InOut parameters	NA
Return	Std_ReturnType
Description	Function that writes PduGroupTX_State to BswM.

[$Satisfies $SRS_PNC 015] [$Satisfies $SRS_PNC 022] [$Satisfies $SRS_PNC 030]

4.11 Detailed Design

4.11.1 Macros

For macros detailed design please refer to Doxygen documentation. Main pAges are:

- Shutdown macros (static code)
- Deactivation macros (static code)

- Alarm macros (static code)
- Timer macros (static code)
- Configuration Macros (dynamic code)
- Utilities

4.11.2 Data Types

PNC_State

[$SDD_PNC 031]

Name	PNC_State
Type	Definition
Range	SUM_PNC_NO_COMMUNICATION 0x00 SUM_PNC_PREPARE_SLEEP0x01 SUM_PNC_READY_SLEEP 0x02 SUM_PNC_REQUESTED 0x03
Description	Used for Mode Request (PN Request fromSUM_PNC) and Mode Switch(Current state from ComM) States reflect the state transitions identified for partial networks in ComM. Requestor of Partial Network will use following states for Mode Requests for function PNC_BswM_CurrentPNCModeReq <n> • SUM_PNC_NO_COMMUNICATION • SUM_PNC_REQUESTED • Mode Switch Interface • PNC_BswM_CurrentPNCModeUser<n> for individual PN will return the current PN state using all states defined in range.

[$Satisfies $SRS_PNC 045] [$Satisfies $SRS_PNC 018]

PNC_Modes

[$SDD_PNC 044]

Name	PNC_Modes
Type	Enum
Range	PNC_ENABLE_CHECK=0, PNC_MONITOR_ACTIVATION, PNC_ACK_ACTIVATION, PNC_TX_ENABLE,PNC_LOCALLY_ACTIVE, PNC_REMOTELY_ACTIVE, PNC_PREPARE_SLEEP

Name	PNC_Modes
Description	Describes the sub states of PN_IDLE and PN_ACTIVE, which helps in understanding local/remote activation of a PN. Based on the activation criteria, the PNC modes transition. When a PN is inactive (PN<n>_IDLE), SUM_PNC shall be set to PNC_ENABLE_CHECK. Once activation criteria is determined, SUM_PNC shall transition to PNC_MONITOR_ACTIVATION in case of local activation or to PNC_TX_ENABLE in case of remote activation.

[$Satisfies $SRS_PNC 030] [$Satisfies $SRS_PNC 035]

PduGroup_State

[$SDD_PNC 033]

Name	PduGroup_state
Type	Definition
Range	SUM_PDUGRP_STOP 0x00 SUM_PDUGRP_START 0x01
Description	Used for Mode Request Interface PNC_BswMPduGroupSwitchTxReq<n> and PNC_BswMPduGroupSwitchRxReq<n> (to Com) to request for Tx or Rx PDU Group enabling associated with specific Partial networks. The initial value is set as SUM_PDUGRP_STOP.

[$Satisfies $SRS_PNC 047]

ComMChannel_State

[$SDD_PNC 032]

Name	ComMChannel_State
Type	Definition
Range	SUM_COMM_NO_COMMUNICATION 0x00 SUM_COMM_FULL_COM_READY_SLEEP 0x01 SUM_COMM_FULL_COM_NETWORK_REQUESTED 0x02 SUM_COMM_FULL_COMMUNICATION 0x03 SUM_COMM_SILENT_COMMUNICATION 0x04
Description	Used for Mode Switch Interface PNC_ComM_CurrentModeUser <n> to indicate current state of ComM for the desired communication channel. Initial value is set to SUM_COMM_NO_COMMUNICATION

[$Satisfies $SRS_PNC 046]

4.11.3 Structures

Please refer to the following pages of doxygen documentation:

- Struct Const Type
- Struct Type
- Channel Info Type

4.12 Runtime Scenario

4.12.1 Activation by Local ECU when ECU is asleep

[$SDD_PNC 051]
Goal: To understand the partial network Activation by local ECU when the ECU is asleep
Pre-Condition: PNC is initialized, Also RTE and SCHM are initiated prior to PNC
Activities: Follow the numbering to understand the sequence of activity.

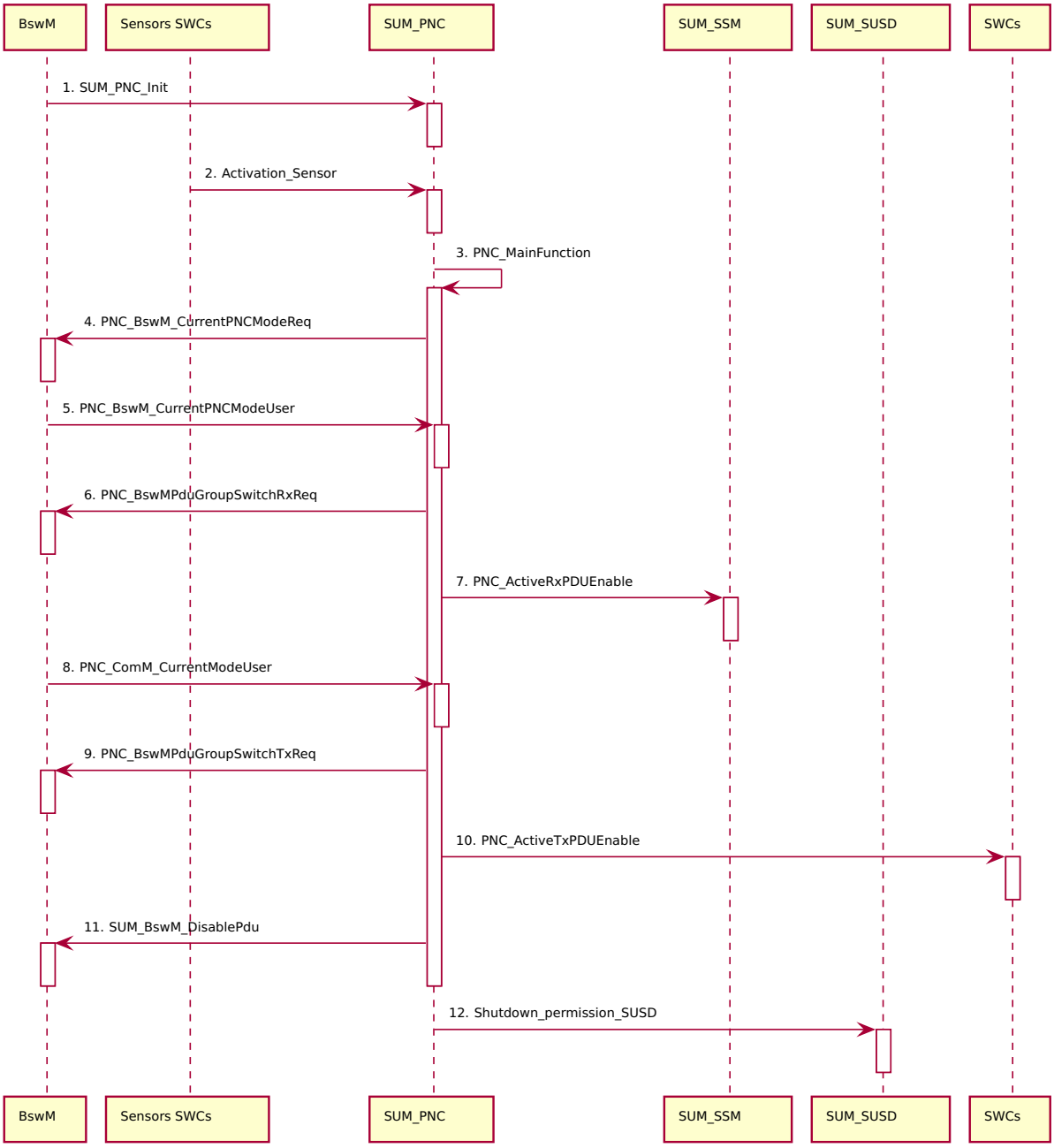


Diagram 4.7

[$Satisfies $SRS_PNC 020] [$Satisfies $SRS_PNC 021] [$Satisfies $SRS_PNC 022] [$Satisfies $SRS_PNC 023] [$Satisfies $SRS_PNC 024]

4.12.2 Activation by Remote ECU when ECU is asleep

[$SDD_PNC 052]

Goal: To understand the partial network Activation by Remote ECU when the ECU is asleep
Pre-Condition: PNC is initialized, Also RTE and SCHM are initiated prior to PNC
Activities: Follow the numbering to understand the sequence of activity.

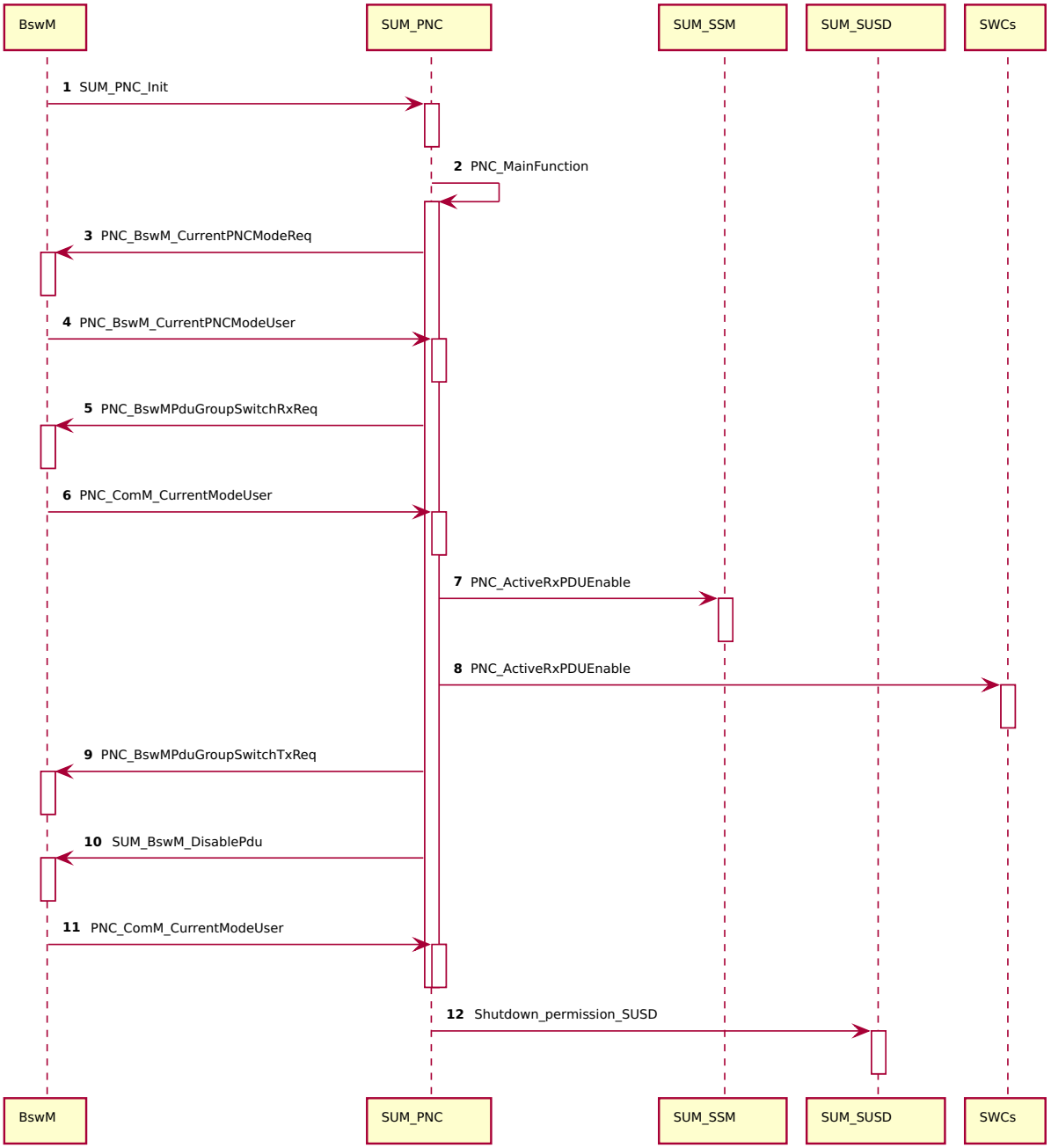


Diagram 4.8

[$Satisfies $SRS_PNC 020] [$Satisfies $SRS_PNC 021] [$Satisfies $SRS_PNC 022] [$Satisfies $SRS_PNC 023] [$Satisfies $SRS_PNC 024]

4.12.3 Activation by Remote ECU when ECU is awake

[$SDD_PNC 053]

Goal: To understand the partial network Activation by remote ECU when the ECU is awake
Pre-Condition: PNC is initialized, Also RTE and SCHM are initiated prior to PNC
Activities: Follow the numbering to understand the sequence of activity.

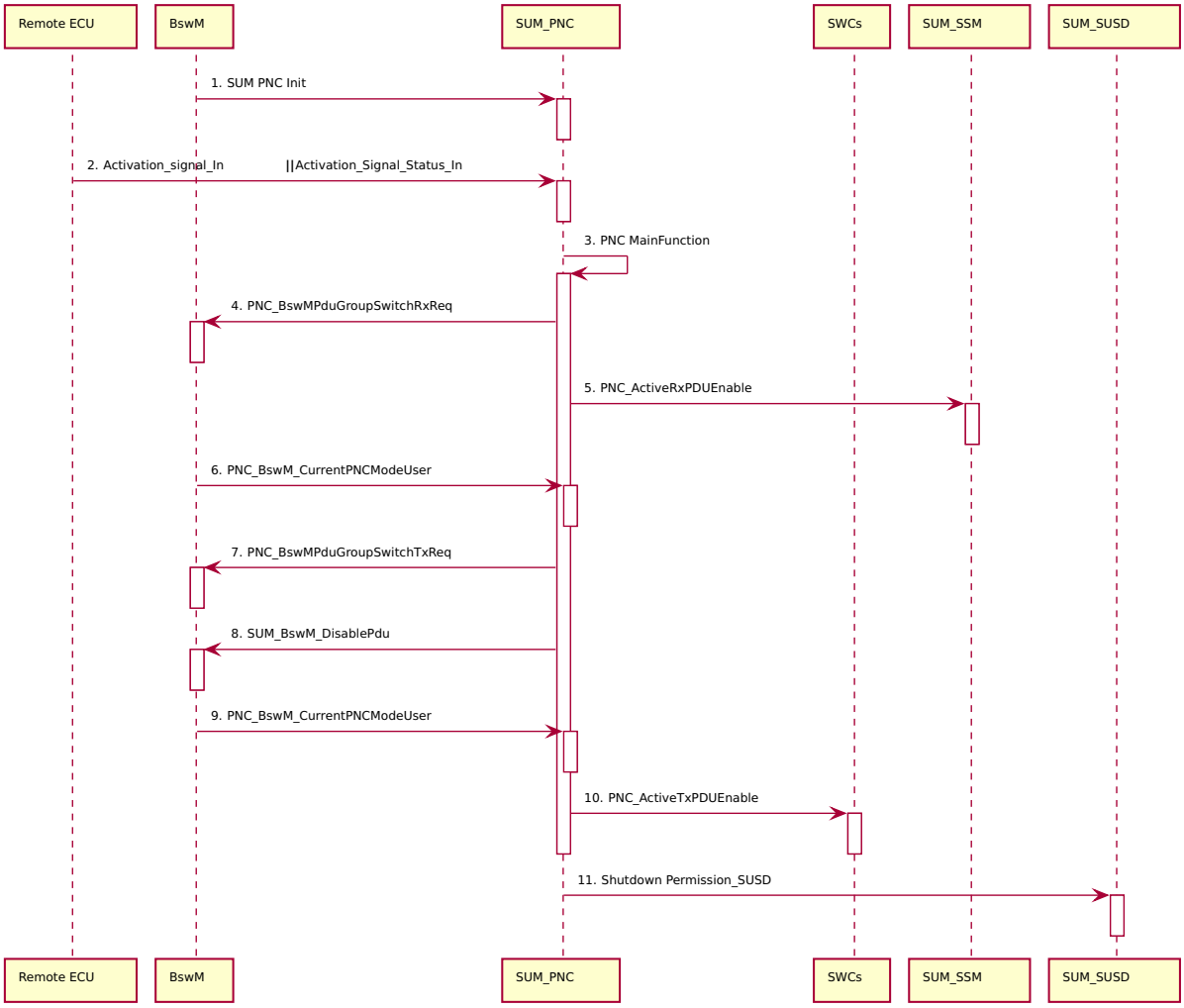


Diagram 4.9

[$Satisfies $SRS_PNC 020] [$Satisfies $SRS_PNC 021] [$Satisfies $SRS_PNC 022] [$Satisfies $SRS_PNC 023] [$Satisfies $SRS_PNC 024]

4.12.4 Activation by Local ECU when ECU is awake

[$SDD_PNC 054]

Goal: To understand the partial network Activation locally when activation criteria is TRUE, when the ECU is awake

Pre-Condition: PNC is initialized, Also RTE and SCHM are initiated prior to PNC

Activities: Follow the numbering to understand the sequence of activity.

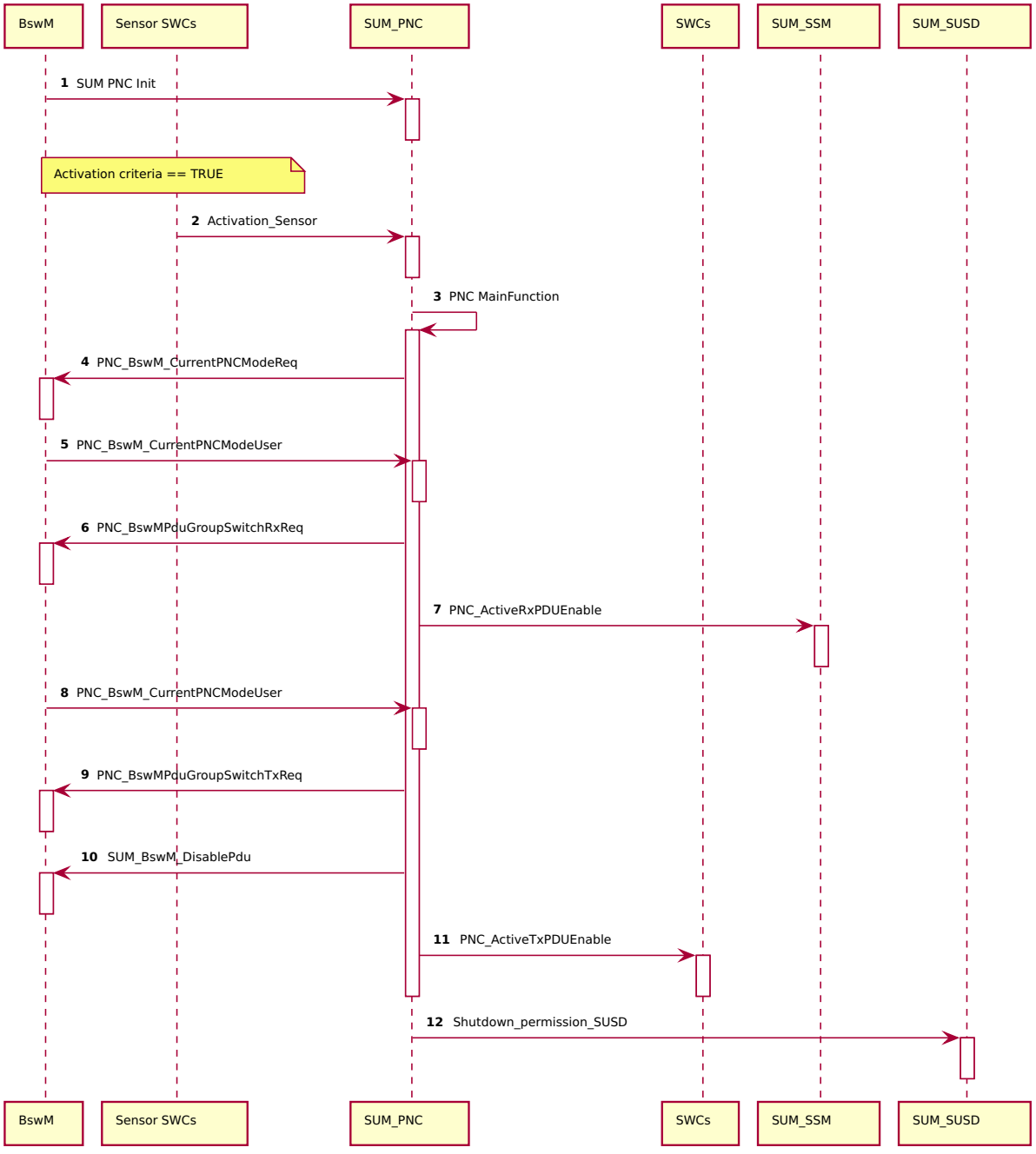


Diagram 4.10

[$Satisfies $SRS_PNC 020] [$Satisfies $SRS_PNC 021] [$Satisfies $SRS_PNC 022] [$Satisfies $SRS_PNC 023] [$Satisfies $SRS_PNC 024]

4.12.5 Remote ECU relinquishes PN activation for a remotely active PN

Goal: To understand the partial network Activation when
Pre-Condition: PNC is initialized, Also RTE and SCHM are initiated prior to PNC
Activities: Follow the numbering to understand the sequence of activity.

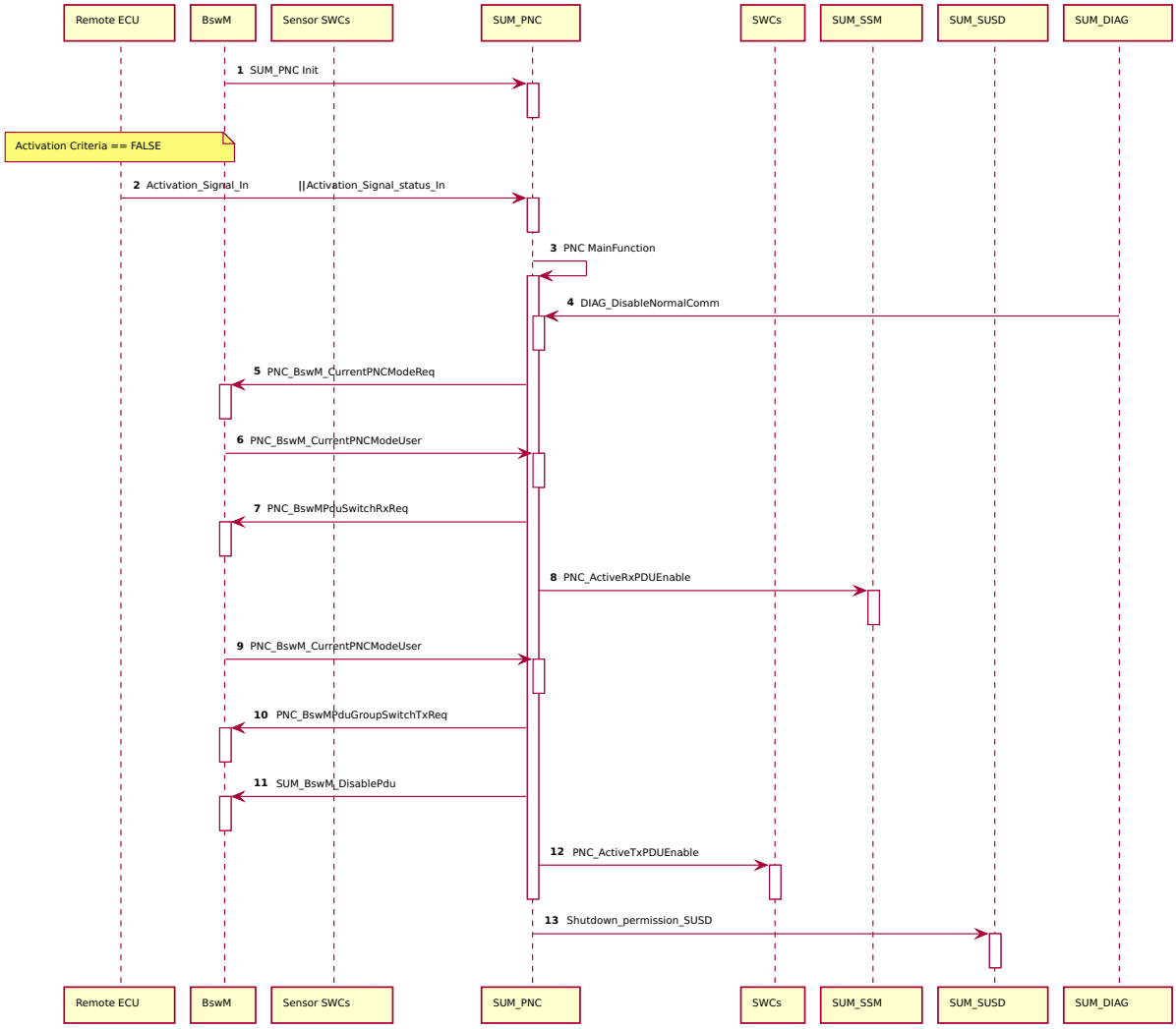


Diagram 4.11

4.12.6 Deactivation Criteria is detected FALSE for a Remotely active PN

[$SDD_PNC 047]

Goal: To understand the behavior when the partial network is in PN<n>_ACTIVE state, and SUM_PNC determines the Activation Criteria for a remotely active PN(PNC_REMOTELY_ACTIVE) is true(and De-activation criteria is FALSE), then SUM_PNC should request local activation.

Pre-Condition:

1. PNC is initialized, Also RTE and SCHM are initiated prior to PNC
2. PN is in already ACTIVE and in Remotely Active state.

Activities: Follow the numbering to understand the sequence of activity.

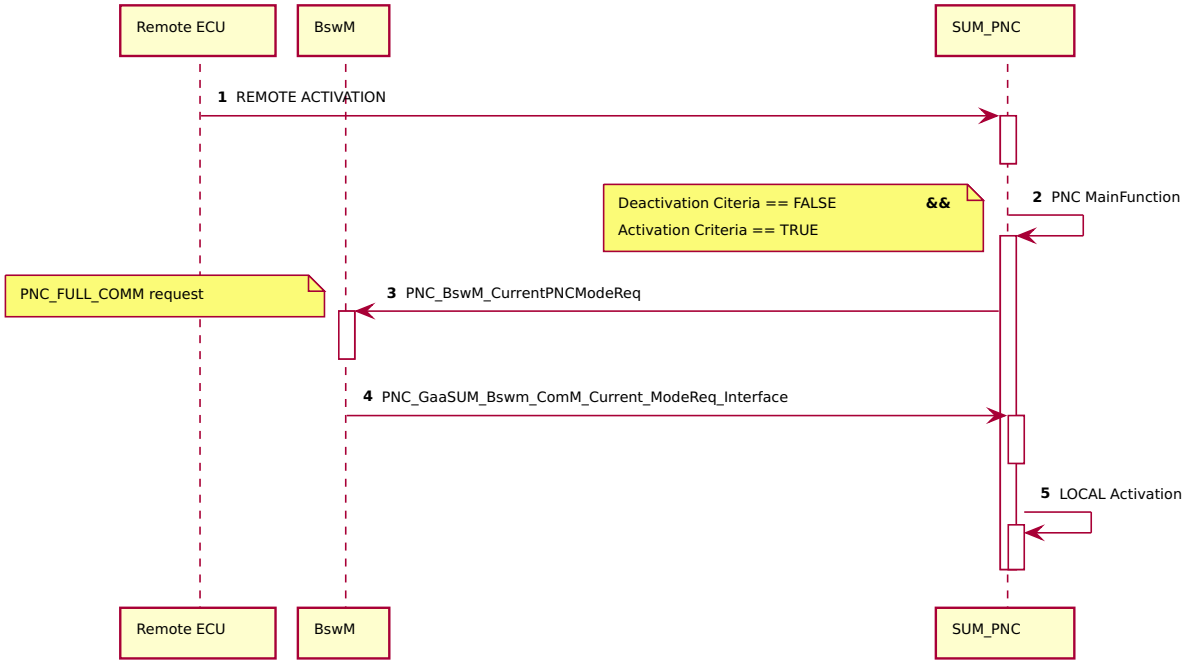


Diagram 4.12

[$Satisfies $SRS_PNC 040]

4.12.7 Deactivation Criteria is detected TRUE for a Locally active PN

[$SDD_PNC 048]

Goal: To understand the behavior when partial network is locally Active state and SUM_PNC determines that the Deactivation criteria is true (and if Activation criteria is False), then SUM_PNC should transition to remotely active

Pre-Condition:

1. PNC is initialized, Also RTE and SCHM are initiated prior to PNC.
2. PN is already active and in locally active state.
3. Repeat Message timer is expired or not started.

Activities: Follow the numbering to understand the sequence of activity.

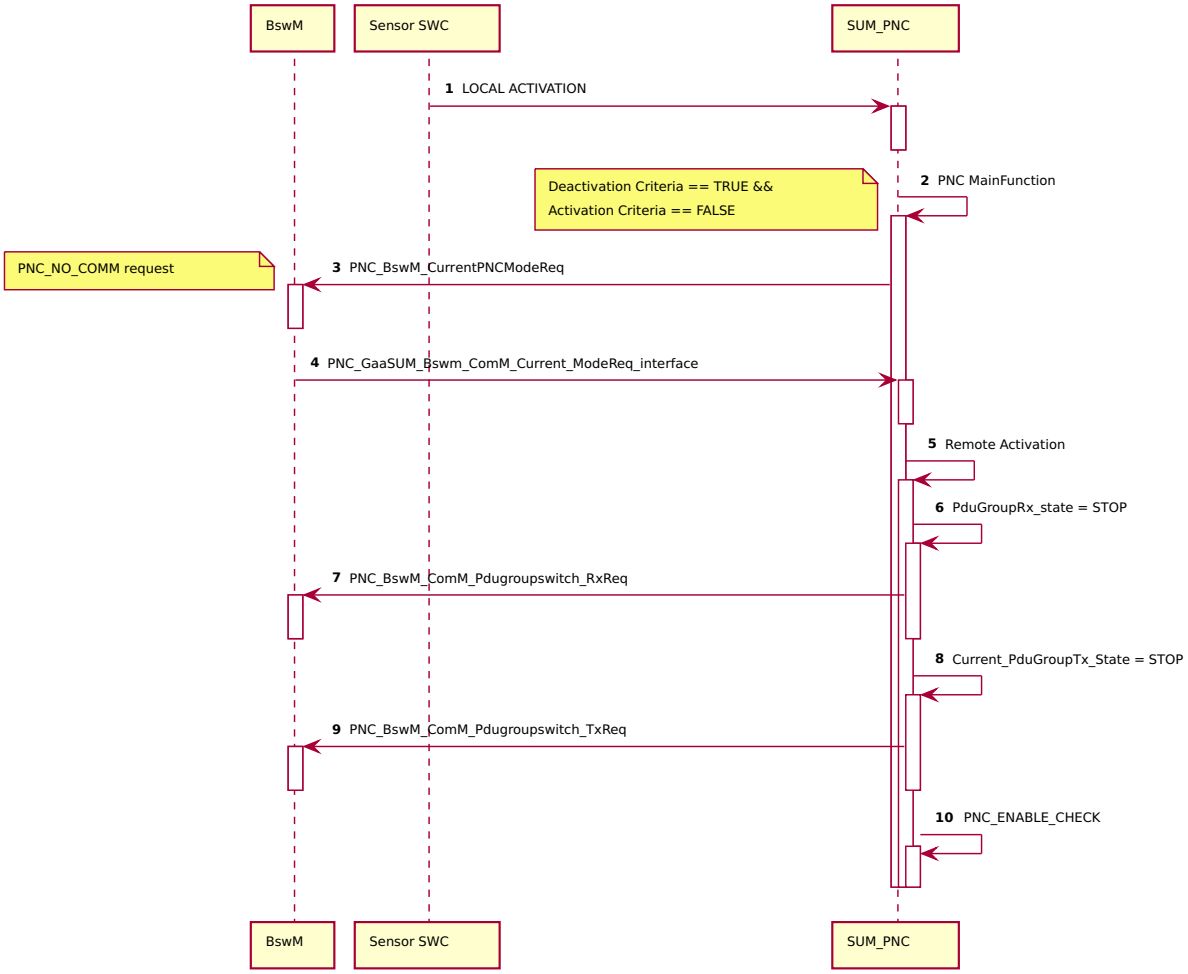


Diagram 4.13

[$Satisfies $SRS_PNC 041]

4.12.8 Sequence for going to Sleep for locally Active PN

[$SDD_PNC 049]

Goal: To understand the sleep procedure when the Partial network is in PNC_LOCALLY_ACTIVE state.
Pre-Condition: PNC is initialized, Also RTE and SCHM are initiated prior to PNC
Activities: Follow the numbering to understand the sequence of activity.

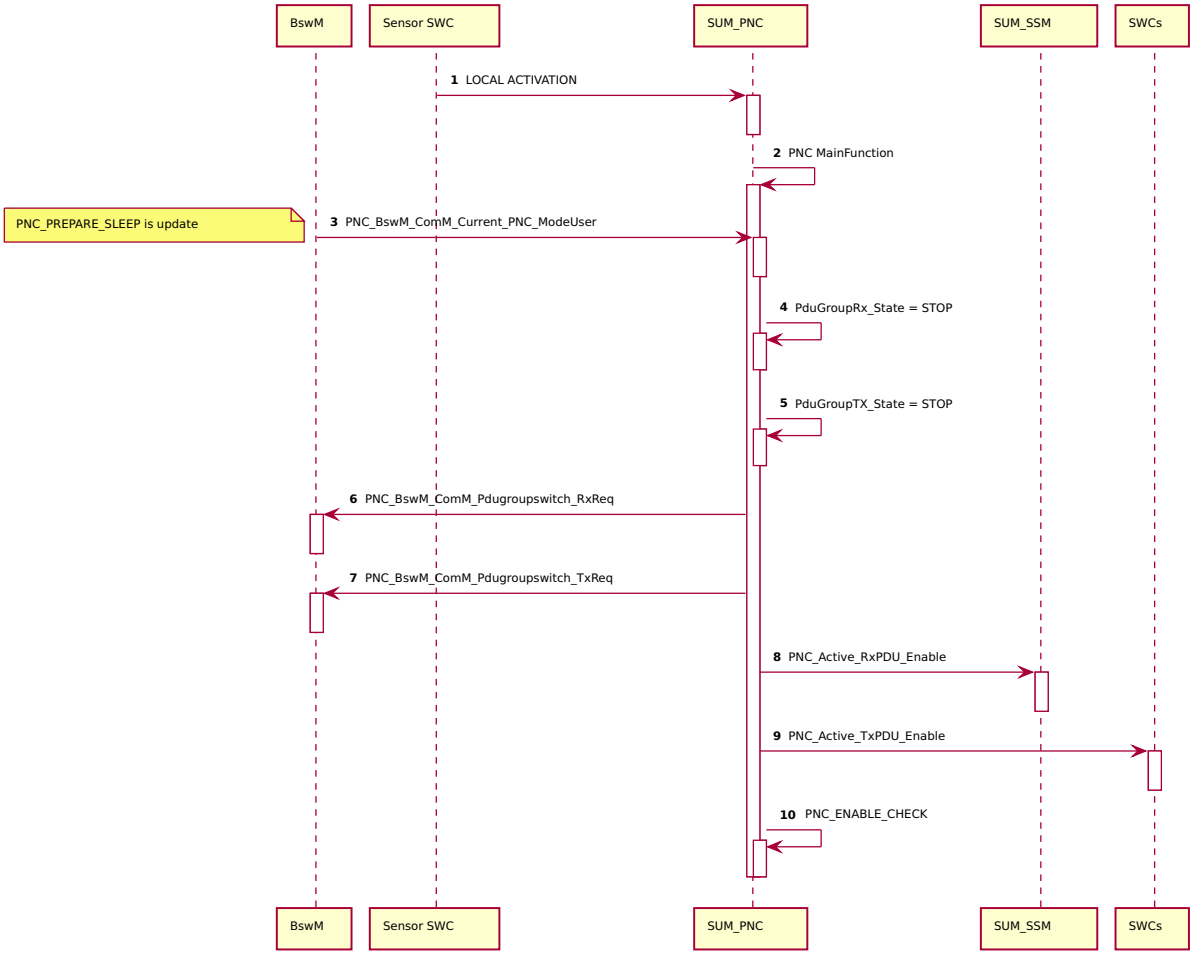


Diagram 4.14

[$Satisfies $SRS_PNC 042]

4.12.9 Sequence for going to Sleep for Remotelly Active PN

[$SDD_PNC 050]

Goal: To understand the Sleep procedure when the PN is in PNC_REMOTELY_ACTIVE state.
Pre-Condition: PNC is initialized, Also RTE and SCHM are initiated prior to PNC
Activities: Follow the numbering to understand the sequence of activity.

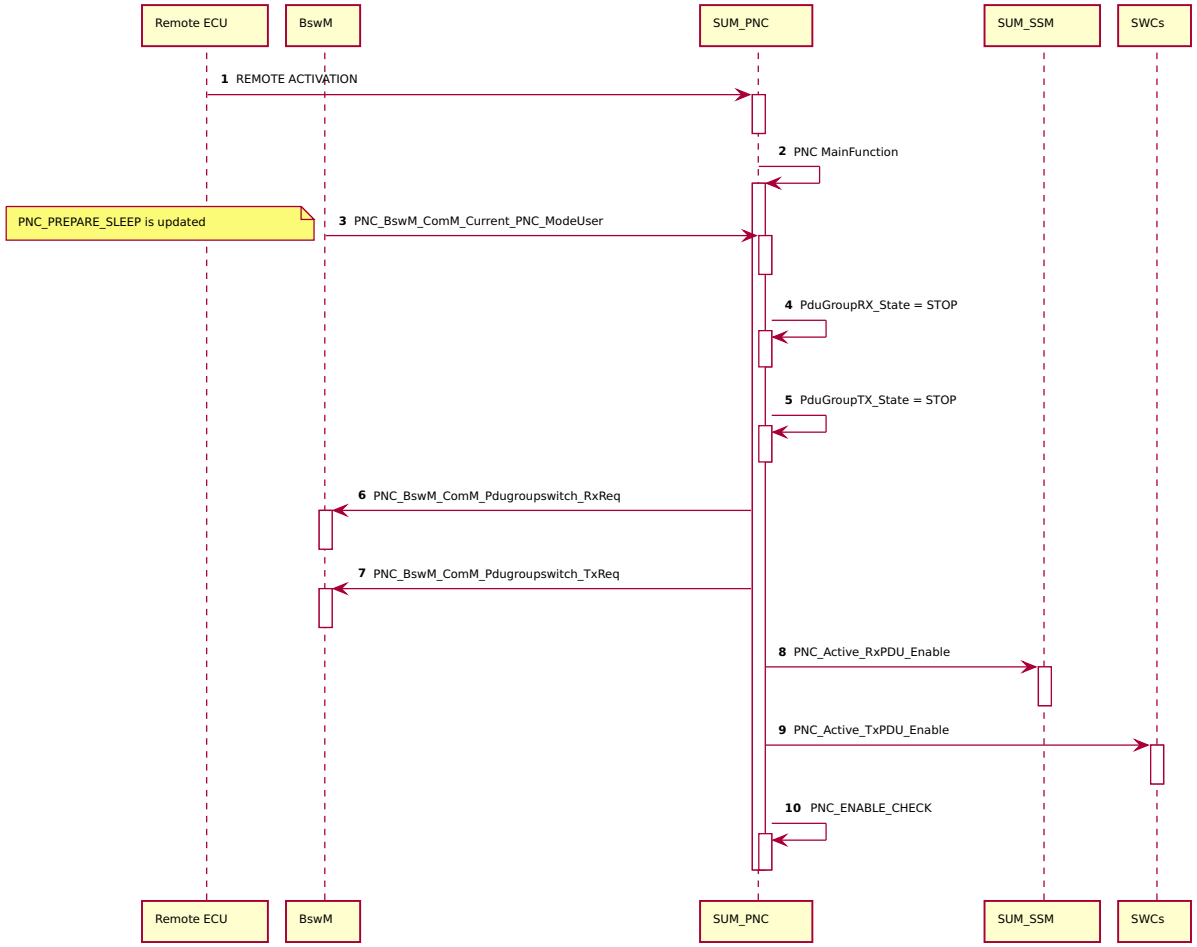


Diagram 4.15

[$Satisfies $SRS_PNC 042]

5 Deployment View

5.1 Resource limits

- **ROM**

Type	Size
.rodata	10 bytes
.vletext (Code)	6560 bytes

- **RAM**

Type	Size
.data	217 bytes
.bss	75 byte

5.2 Hardware Resources

<N/A>

5.2.1 Interrupts

<N/A>

5.2.2 Communication Channels

Configure ComM channel and corresponding PN related information in BSW Ecuc values Arxml. Reference of ComM configuration shall be used in SUM_PNC module configuration. Refer section 3.2 for details.

5.2.3 Digital and Analog I/O

<N/A>

5.2.4 Watchdog

<N/A>

5.2.5 Resource N

<N/A>

5.3 External Hardware Dependencies

<N/A>